

EEBus UC Technical Specification

Monitoring of Grid Connection Point

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1 Scope of the document

This document describes the Use Case "Monitoring of Grid Connection Point" (short-name: MGCP). Chapter 2 specifies the High-Level Use Case. Chapter 3 details the technical solution for SPINE for this Use Case. Within this document, a top-down approach is used to derive the requirements for the technical solution from the High-Level description.

1.1 References

1.1.1 EEBUS documents

[ProtocolSpecification] EEBus_SPINE_TS_ProtocolSpecification.pdf

[ResourceSpecification] EEBus_SPINE_TS_ResourceSpecification.pdf

[SHIP] EEBus_SHIP_TS_Specification_v1.0.1.pdf

1.1.2 Normative references

[RFC2119] IETF RFC 2119: 1997, Key words for use in RFCs to indicate requirement levels
Please see section 1.3.1 for details.

1.2 Terms and definitions

AC

Abbreviation for alternating current

Actor

An Actor models a role within a Use Case definition (e.g. an energy manager or a heat pump).

CEM

Abbreviation for Customer Energy Manager. The CEM is an energy manager located at the building of the user or in a cloud application. The energy manager enables energy-optimized operation of the connected devices by harmonising energy demand and availability.

Configuration Appliance

The Actor Configuration Appliance configures particular data of another Actor.

MGCP

Monitoring of Grid Connection Point (short name of this Use Case)

Monitoring Appliance

The Actor Monitoring Appliance evaluates particular data of another Actor.

pct

Abbreviation for percentage

PLF

Abbreviation for power limitation factor

177 Polling

178 Mechanism where data is requested by the client periodically. Typically, polling needs to be
179 repeated frequently by the client to increase the probability to notice all changes from a data server
180 in time.

181 PV

182 Abbreviation for photovoltaic (system)

183 Scenario

184 Part of a Use Case. Splitting a Use Case into Scenarios helps readers understand the Use Case more
185 quickly. Some Scenarios are mandatory for a Use Case, whereas others may be recommended or
186 optional.

187 Specialization

188 Reusable data collection for a specific functionality.

189 SPINE

190 Smart Premises Interoperable Neutral-message Exchange: Technical Specification of EEBus Initiative
191 e.V.

192

193 1.3 Requirements**194 1.3.1 Requirements wording**

195 The following keywords are used:

- 196 - SHALL
- 197 - SHALL NOT
- 198 - SHOULD
- 199 - SHOULD NOT
- 200 - MAY

201 Note: They apply only if written in capital letters.

202 For the meaning of the keywords, please refer to [RFC2119].

203

204 1.3.2 Mapping of High-Level requirements

205 Within the High-Level Use Case description, the following abbreviation is used:

206 [MGCP-xyz]

207 e.g.: [MGCP-007]

208 The abbreviation is used to mark High-Level requirements or rules of this Use Case with a unique
209 number xyz. These requirements are referenced throughout the technical solution to show how each
210 High-Level requirement is realized in the technical part.

211

2 High-Level description

2.1 Introduction

The Monitoring Appliance monitors electrical measurands of the Grid Connection Point. A Grid Connection Point is defined as the point where the public electricity grid is connected to the internal grid of the building. The public grid as well as the internal grid are AC electricity grids. The Actor "Grid Connection Point" represents the measured values of the physical Electricity Grid Connection Point. The measurement is typically performed by an electrical smart meter or submeter.

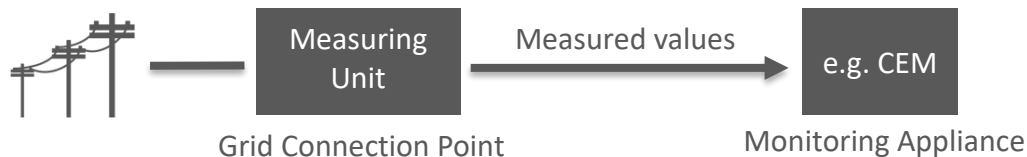


Figure 1: High-Level Use Case functionality overview

Added value: This general-purpose Use Case is used for visualization or monitoring of electrical smart meter or submeter values at the Grid Connection Point. This is for example useful if PV power exceeds the maximum feed-in power. In this case (e.g.) an energy manager can offer energy to the connected devices at no cost with zero emission foot print by not power curtailing the PV system. In addition, the autarky rate will be increased.

2.2 Detailed background information

This Use Case describes systems where electrical consumers and electrical producers in the building of the user are connected to the internal electrical grid of the building.

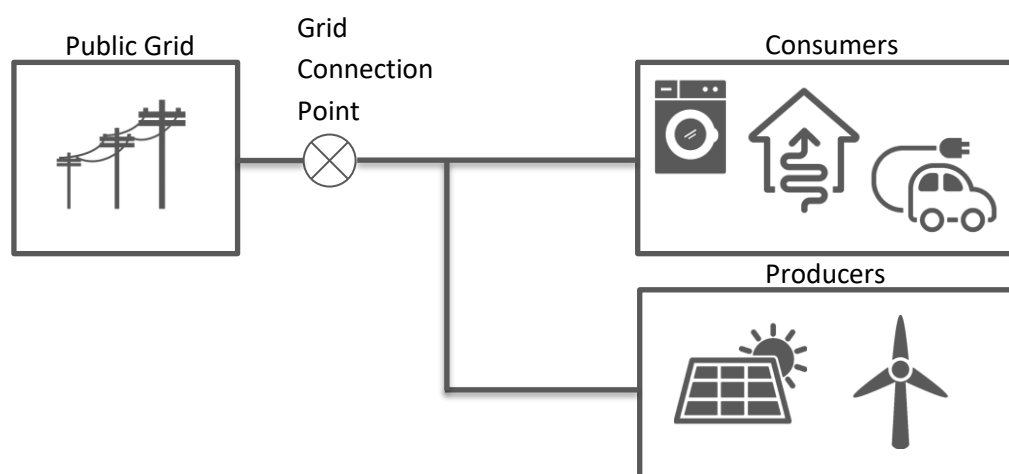


Figure 2: Location of the Grid Connection Point

234 The Electricity Grid Connection Point provides some or all following data points to the Monitoring
235 Appliance:

- 236 - PV feed-in power limitation factor
- 237 - Momentary power consumption/production
- 238 - Total feed-in energy
- 239 - Total consumed energy
- 240 - Momentary current consumption/production (phase-specific)
- 241 - Voltage (phase-specific)
- 242 - Frequency

243

244 **2.3 Actors**

245 **2.3.1 Monitoring Appliance**

246 The Actor Monitoring Appliance (e.g. a CEM) may manage the home's appliances in such a way that
247 lets the customer benefit from the energy optimized operation of the appliances or just visualize the
248 values for the customer. In this Use Case the Monitoring Appliance retrieves power, energy, current
249 or other measurement data provided by the Grid Connection Point's electrical smart meter or
250 submeter.

251

252 **2.3.2 Grid Connection Point**

253 This Actor is a Grid Connection Point, representing the point where the public electricity grid is
254 connected to the internal grid of the building. An electrical smart meter or submeter provides
255 relevant measurement data to the home's energy manager.

256

2.4 Scenarios

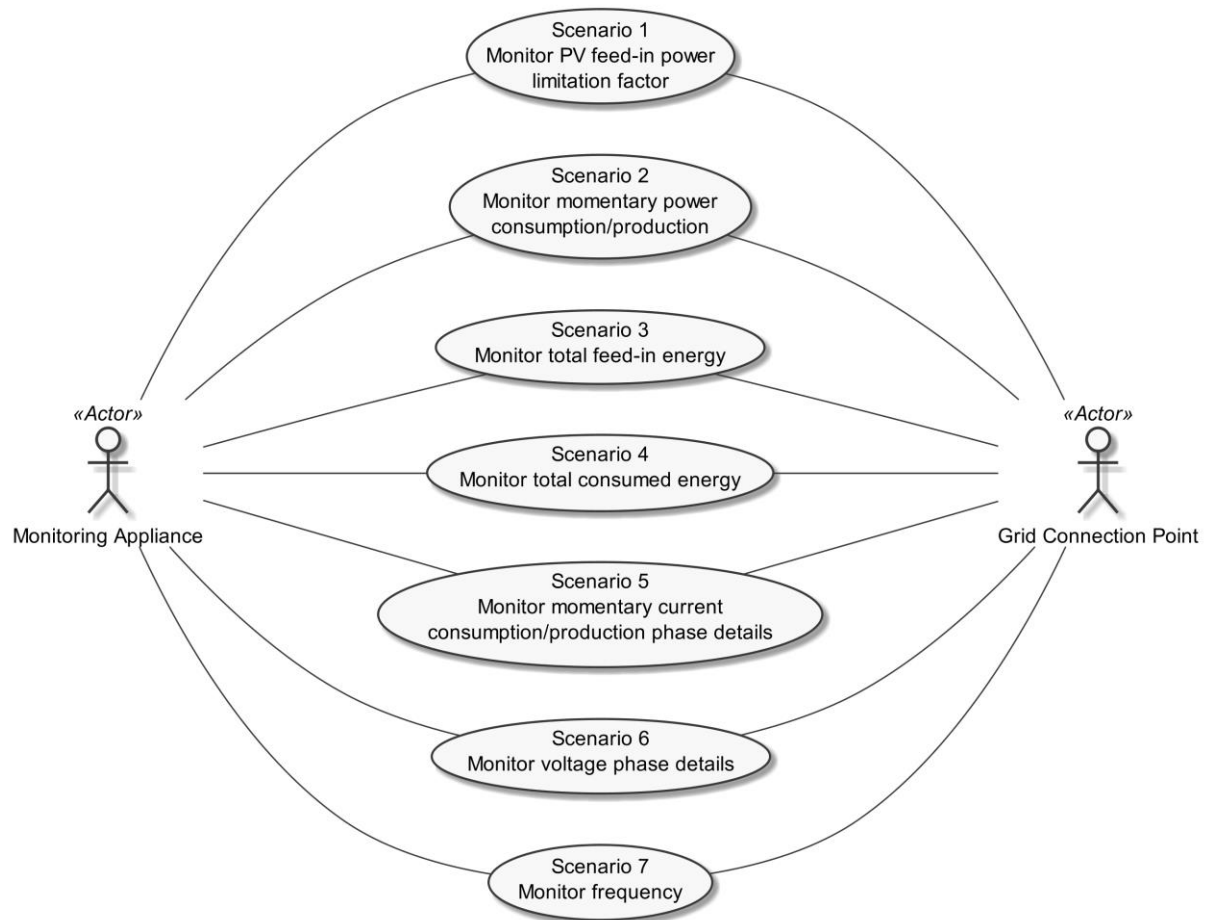


Figure 3: Scenario overview

Scenario number	Scenario name	Monitoring Appliance	Grid Connection Point
1	Monitor PV feed-in power limitation factor	O	O
2	Monitor momentary power consumption/production	R	M
3	Monitor total feed-in energy	O	M
4	Monitor total consumed energy	O	M
5	Monitor momentary current consumption/production phase details	O	R
6	Monitor voltage phase details	O	O
7	Monitor frequency	O	O

Table 1: Scenario implementation requirement for Actors

At least one of the Scenarios 2, 3 or 4 SHALL be supported by the Actor Monitoring Appliance.

Note: Scenario 5 is needed for phase-specific energy management only.

2.4.1 Scenario 1 - Monitor PV feed-in power limitation factor

2.4.1.1 Description

The Monitoring Appliance wants to be informed about the maximum power value allowed to be fed into the public electricity grid from the internal grid of the premises, as percental value of the cumulated nominal peak power of all electricity producing PV systems within the building ([MGCP-011]).

The following equation applies: $P_{PV, \text{feed-in}} \leq PLF_{PV, \text{feed-in, max, pct}} * \sum P_{PV, AC, \text{nom}}$

With:

- $P_{PV, \text{feed-in}}$ = Maximum allowed feed-in power
- $PLF_{PV, \text{feed-in, max, pct}}$ = This Scenario's value
- $P_{PV, AC, \text{nom}}$ = Cumulated nominal peak power of all PV systems within the building

Data point name	Data point description	Support indication	High-Level requirement
PV Feed-In Power Limitation Factor	Maximum allowed feed-in power as percental value of the cumulated nominal peak power of all electricity producing PV systems within the building.	M	[MGCP-011]

Table 2: Scenario 1 - Monitor PV feed-in power limitation factor - Data point list

2.4.1.2 Conditions

Triggering Event:

The Actor Monitoring Appliance is interested in the PV feed-in power limitation factor of the Actor Grid Connection Point.

Pre-condition:

The Actor Monitoring Appliance does not know the PV feed-in power limitation factor of the Actor Grid Connection Point.

The PV feed-in power limitation factor ($PLF_{PV, \text{feed-in, max, pct}}$) and the nominal peak power of the individual electricity producing PV system ($P_{PV, AC, \text{nom}}$) SHALL have been configured by the installer or the user according to the local regulations.

Note: $P_{PV, AC, \text{nom}}$ is not part of this Use Case.

Post-condition:

The Actor Monitoring Appliance knows the PV feed-in power limitation factor of the Actor Grid Connection Point.

2.4.2 Scenario 2 - Monitor momentary power consumption/production

2.4.2.1 Description

The Monitoring Appliance wants to be informed about the momentary value ([MGCP-004]) of the electrical active power that is consumed from the public grid by local appliances or electrical power that is fed into the public grid from the internal grid of the local premises ([MGCP-021]). This value is the sum of all connected phases ([MGCP-005]).

Data point name	Data point description	Support indication	High-Level requirement
Total Active Power	The total active power at the Grid Connection Point.	M	[MGCP-021]

Table 3: Scenario 2 - Monitor momentary power consumption/production - Data point list

2.4.2.2 Conditions

Triggering Event:

The Actor Monitoring Appliance is interested in the momentary power consumption/production value of the Actor Grid Connection Point.

Pre-condition:

The Actor Monitoring Appliance does not know the momentary power consumption/production value of the Actor Grid Connection Point.

Post-condition:

The Actor Monitoring Appliance knows the momentary power consumption/production value of the Actor Grid Connection Point.

2.4.3 Scenario 3 - Monitor total feed-in energy

2.4.3.1 Description

The Monitoring Appliance wants to be informed about the cumulated electrical energy that was fed into the public grid from the internal grid of the premises ([MGCP-031]). This value is the sum of all connected phases ([MGCP-005]). The total feed-in energy only counts energy fed into the grid and is not reduced by energy consumed from the grid ([MGCP-007]). Only the latest value of the cumulated electrical energy is considered ([MGCP-006]).

Data point name	Data point description	Support indication	High-Level requirement
Total Grid Feed-In Energy	The total energy that was fed into the public electricity grid.	M	[MGCP-031]

Table 4: Scenario 3 - Monitor total feed-in energy - Data point list

2.4.3.2 Conditions

Triggering Event:

The Actor Monitoring Appliance is interested in the total feed-in energy of the Actor Grid Connection Point.

Pre-condition:

The Actor Monitoring Appliance does not know the total feed-in energy of the Actor Grid Connection Point.

Post-condition:

The Actor Monitoring Appliance knows the total feed-in energy of the Actor Grid Connection Point.

2.4.4 Scenario 4 - Monitor total consumed energy**2.4.4.1 Description**

The Monitoring Appliance wants to be informed about the cumulated electrical energy that was consumed from the public grid by local appliances ([MGCP-041]). This value is the sum of all connected phases ([MGCP-005]). The total consumed energy only counts energy consumed from the grid and is not reduced by energy fed into the grid ([MGCP-008]). Only the latest value of the cumulated electrical energy is considered ([MGCP-006]).

Data point name	Data point description	Support indication	High-Level requirement
Total Grid Consumed Energy	The total energy that was consumed from the public electricity grid.	M	[MGCP-041]

Table 5: Scenario 4 - Monitor total consumed energy - Data point list

2.4.4.2 Conditions**Triggering Event:**

The Actor Monitoring Appliance is interested in the total consumed energy of the Actor Grid Connection Point.

Pre-condition:

The Actor Monitoring Appliance does not know the total consumed energy of the Actor Grid Connection Point.

Post-condition:

The Actor Monitoring Appliance knows the total consumed energy of the Actor Grid Connection Point.

2.4.5 Scenario 5 - Monitor momentary current consumption/production phase details**2.4.5.1 Description**

The Monitoring Appliance wants to be informed about the momentary value ([MGCP-004]) of the current that is consumed from the public grid by local appliances or current that is fed into the public grid from the internal grid of the local premises ([MGCP-051]). The value(s) are phase-individual (not the sum of all phases). Only the active current is considered in this Use Case ([MGCP-050/1]). Only rms values are considered ([MGCP-050/2]).

Data point name	Data point description	Support indication	High-Level requirement
Phase-Specific AC Current	The phase-specific active AC current: - Current on phase A ○ [MGCP-051/1] - Current on phase B ○ [MGCP-051/2] - Current on phase C ○ [MGCP-051/3]	M	[MGCP-051]

Table 6: Scenario 5 - Monitor momentary current consumption/production phase details - Data point list

2.4.5.2 Conditions

Triggering Event:

The Actor Monitoring Appliance is interested in the momentary current consumption/production value(s) of the Actor Grid Connection Point.

Pre-condition:

The Actor Monitoring Appliance does not know the momentary current consumption/production value(s) of the Actor Grid Connection Point.

Post-condition:

The Actor Monitoring Appliance knows the momentary current consumption/production value(s) of the Actor Grid Connection Point.

2.4.6 Scenario 6 - Monitor voltage phase details

2.4.6.1 Description

The actual phase-specific AC voltage values of the Actor Grid Connection Point are provided by this Scenario ([MGCP-061]). Depending on the number of connected phases^{*1}, the Grid Connection Point provides a different amount of individual values.

Data point name	Data point description [High-Level requirement]	Support indication	High-Level requirement
Phase-Specific AC Voltage	Voltage between phase A and neutral [MGCP-061/1].	R ^{*1}	[MGCP-061]
	Voltage between phase B and neutral [MGCP-061/2].	R ^{*1}	
	Voltage between phase C and neutral [MGCP-061/3].	R ^{*1}	
	Voltage between phase A and phase B [MGCP-061/4].	O ^{*1}	
	Voltage between phase B and phase C [MGCP-061/5].	O ^{*1}	
	Voltage between phase C and phase A [MGCP-061/6].	O ^{*1}	

Table 7: Scenario 6 - Monitor voltage phase details - Data point list

^{*1}: Only values related to the connected phases SHALL be delivered.

Note: If this Scenario is supported, at least one of the values stated above SHALL be supported.

2.4.6.2 Conditions

Triggering Event:

The Actor Monitoring Appliance is interested in the phase-specific AC voltage values of the Actor Grid Connection Point.

Pre-condition:

The Actor Monitoring Appliance does not know the phase-specific AC voltage values of the Actor Grid Connection Point.

Post-condition:

The Actor Monitoring Appliance knows the phase-specific AC voltage values of the Actor Grid Connection Point.

2.4.7 Scenario 7 - Monitor frequency

2.4.7.1 Description

The frequency at the Grid Connection Point is provided by this Scenario ([MGCP-071]).

Data point name	Data point description	Support indication	High-Level requirement
AC Frequency	Frequency, the Grid Connection Point measures.	M	[MGCP-071]

Table 8: Scenario 7 - Monitor frequency - Data point list

2.4.7.2 Conditions

Triggering Event:

The Actor Monitoring Appliance is interested in the AC frequency of the Actor Grid Connection Point.

Pre-condition:

The Actor Monitoring Appliance does not know the AC frequency of the Actor Grid Connection Point.

Post-condition:

The Actor Monitoring Appliance knows the AC frequency of the Actor Grid Connection Point.

2.5 Dependencies to other Use Cases

None.

2.6 Further information and rules

2.6.1 Sign conventions

Throughout the whole Use Case the "load convention" (i.e. "passive sign convention") is applied for measurement values ([MGCP-001]). This means measured electrical current, active power and exchanged energy are expressed with positive values in case of energy consumption whereas negative values are used in case of energy production. This holds valid for the following values of this Use Case:

- Total Active Power ([MGCP-021])
- Total Grid Feed-In Energy ([MGCP-031])
- Total Grid Consumed Energy ([MGCP-041])
- Phase-Specific AC Current ([MGCP-051])

The Grid Connection Point's voltages ([MGCP-061]) are measured independent of the energy direction ([MGCP-002]).

2.6.2 Value state rules

Each Grid Connection Point's measurement value has a state that defines whether the value is correct or is to be ignored by the Monitoring Appliance ([MGCP-003]):

- normal: value correct
- out of range: value is out of range and SHALL be ignored by the Monitoring Appliance
- error: value is erroneous and SHALL be ignored by the Monitoring Appliance.

2.7 Assumptions and Prerequisites

- The Grid Connection Point is connected to an electrical smart meter or submeter which provides the relevant power, energy or current data for this Use Case.
- The Grid Connection Point represents a single connection from the building to the public electricity grid.

3 Technical SPINE solution

3.1 General rules and information

3.1.1 Underlying technology documents

This technical solution relies on the SPINE Resources Specification version 1.1.0 [ResourceSpecification].

For interoperable connectivity this technical solution relies on:

- SPINE Protocol Specification version 1.1.0 [ProtocolSpecification] as application protocol.
- SHIP Specification version 1.0.1 [SHIP] as transport protocol.

3.1.2 Use Case discovery rules

Use Case discovery SHOULD be supported by each Actor. If Use Case discovery is supported the following rules SHALL apply:

- The string content for the Element "nodeManagementUseCaseData. useCaseInformation. useCaseSupport. useCaseName" within the Use Case discovery (please refer to [ProtocolSpecification]) SHALL be "monitoringOfGridConnectionPoint". The string content SHALL only be defined by this Use Case (regardless of the Use Case version).
- The string content of the Element "nodeManagementUseCaseData. useCaseInformation. actor" within the Use Case discovery (please refer to [ProtocolSpecification]) SHALL be set to the according value stated within the corresponding Actor's section.
- An Actor A that is implemented to support this Use Case specification SHALL set the Element "nodeManagementUseCaseData. useCaseInformation. useCaseSupport. useCaseVersion" within the Use Case discovery (please refer to [ProtocolSpecification]) to "1.0.0".
- If an Actor A supports multiple versions of this Use Case with the same major version number, only the highest one SHOULD be set within the Use Case discovery.
- If an Actor A finds a proper counterpart Actor B for this Use Case that supports multiple versions of this Use Case with the same major version number as supported by Actor A, the Actor A SHOULD evaluate from these versions of Actor B only the highest version number.
- If an Actor A supports multiple versions of this Use Case with different major version numbers, for each major version number only the highest version number SHOULD be set within the Use Case discovery.
- If an Actor A finds a proper counterpart Actor B for this Use Case that supports only versions with a major version number not implemented by Actor A, it still might be possible to run the Use Case or parts of the Use Case. Therefore, the Actor A should try to evaluate the Actor B as a valid partner for this Use Case.

3.1.3 Rules for "Content of Specialization..." tables and "Content of Function..." tables

3.1.3.1 General presence indication definitions

Abbreviations for the presence indication of Elements listed in the tables are defined as follows:

Abbreviation	Meaning	Link to requirement keywords
M	Mandatory	SHALL
R	Recommended	SHOULD
O	Optional	MAY

Table 9: Presence indication description

An Actor MAY support Elements that are not listed in the tables. However, another Actor MAY ignore these Elements.

The presence indications "M", "R" and "O" are always meant relative to the respective parent Element. I.e. if a parent Element is optional ("O") and a child is mandatory ("M") the child Element can only be present if the parent Element is present as well.

Note: The indications and the aforementioned rules apply for "complete messages" (so-called "full function exchange", please refer to [ProtocolSpecification]). In contrast, the so-called "restricted function exchange" is designed to permit exchange of specific excerpts of data, i.e. fewer Elements than potentially available from the data owner (partially even not all "mandatory" Elements).

3.1.3.2 Presence indications for "Content of Specialization..." tables

This section only defines rules for the client side.

Elements that are marked with "M" SHALL be supported by the client in case of readable as well as writeable data. This Element may be optional on the server side.

The following applies for readable data that is exchanged in a "read/reply" or "notify" operation:

- "R" means that the data SHOULD be supported by the client. In other words: If the server responds with the according Element, the client SHOULD be able to interpret the according Elements.
- "O" means that the data MAY be supported by the client. In other words: If the server responds with the according Element, the client MAY be able to interpret the according Elements.

The following applies for writeable data that is exchanged in a "write" operation:

- "R" means that the data SHOULD be written by the client.
- "O" means that the data MAY be written by the client.
- "F" means that the data SHALL NOT be written by the client.

The following applies for Elements that are not listed in the Actor section:

- In case of a received "reply" message: The client MAY ignore the Element.
- In case of a "write" operation to be created: The client MAY set the Element but SHALL consider that the server may ignore the Element.
- In case of a received "notify" message: The client MAY ignore the Element.

M, R or O may be combined with the suffix "(event)" to express that a supported Element or value only has to be supported during a certain event and hence does not need to be present at all times. If

505 the event is not active the Element may be omitted or another value may be set. In most cases a
506 High-Level requirement reference for the event is given in the rules column.

507 In some cases, an Element may have a double presence indication, separated by a colon. Examples:

- 508 - M, O
- 509 - M \W, O \W

510 The description (rules) of the Element details in such a case the presence requirement. A typical
511 example is a list-based Element where the "first" (or "last") Element has a different presence
512 requirement than the other list Elements.

513

514 **3.1.3.3 Presence indications for "Content of Function..." tables**

515 This section only defines rules for the server side.

516 Elements that are marked with "M" SHALL be supported by the server in case of readable as well as
517 writeable data. In case of writeable data (marked with "M \W") the server does not need to set the
518 Element, because the Element is set only by the client.

519 The following applies for readable data that is exchanged in a "read/reply" or "notify" operation:

- 520 - "R" means that the data SHOULD be provided by the server.
- 521 - "O" means that the data MAY be provided by the server.
- 522 - "F" means that the data SHALL NOT be provided by the server.

523 The following applies for writeable data that is exchanged in a "write" operation:

- 524 - "R" means that the data SHOULD be supported. In other words: If the client writes the
525 Element, the server SHOULD accept those messages and the contained Elements.
- 526 - "O" means that the data MAY be supported. In other words: If the client writes the Element,
527 the server MAY accept those messages and the contained Elements.

528 The following applies for Elements that are not listed in the Actor section:

- 529 - In case of a received "read" request: The according Element MAY be set in the reply.
- 530 - In case of a received "write" operation: The server MAY ignore the Element.
- 531 - In case of a "notify" operation to be created: The server MAY set the Element.

532 Note: The server will only accept write operations if the result fulfils the server Function
533 requirements (permitted values, e.g.). Write operations on Elements that are not writeable MAY
534 result in an error message.

535 M, R or O may be combined with the suffix "(event)" to express that a supported Element or value
536 only has to be supported during a certain event and hence does not need to be present at all times. If
537 the event is not active the Element may be omitted or another value may be set. In most cases a
538 High-Level requirement reference for the event is given in the rules column.

539 In some cases, an Element may have a double presence indication, separated by a colon. Examples:

- 540 - M, O

541 - M \W, O \W

542 The description (rules) of the Element details in such a case the presence requirement. A typical
543 example is a list-based Element where the "first" (or "last") Element has a different presence
544 requirement than the other list Elements.

545

546 **3.1.3.4 Cardinality indications on Elements and list items**

547 A cardinality indication on an Element or list item expresses constraints on the number of
548 occurrences of a given Element or data set. In this section we use "X" as representation for such an
549 Element or data set. Furthermore, "a" and "b" represent constraints. The following rules apply for
550 the occurrence of "X" and its content related to a specific Scenario (see note underneath the list):

- 551 1. X
552 No cardinality indication.
- 553 2. X (a..b)
554 This means "X" SHALL occur at least "a" times and at maximum "b" times.
- 555 3. X (a..)
556 This means "X" SHALL occur at least "a" times and MAY occur more than "a" times.
- 557 4. X (..b)
558 This means "X" SHALL occur at maximum "b" times and MAY occur less than "b" times (even
559 zero occurrences are permissive).

560 Note: These rules apply only under consideration of presence indications and with regards to the
561 given Scenario or Function definition for this Use Case.

562 The following table is an example to explain this for two different placements.

Scenario [...]: M/R/O [W]\C]	Element	Value	[High-Level Mapping] Element and value rules
...	...	...	...
2: M \W	xFeatureType. xListData. xData. (1..3)		
2: M \W	xId	<*(1..)>	PRIMARY IDENTIFIER
2: M \W	timePeriod		...
2: M \W	timePeriod. startTime	<xs:duration>	
2: M \W	xSlot. (1..)		
2: M \W	xSlot. xSlotId		...
2: M \W	xSlot. duration	<xs:duration>	...
...	...	...	...

563 Table 10: Example table for cardinality indications on Elements and list items

564 The field

565 xFeatureType. xListData. xData. (1..3)

introduces a data pattern (required Elements and values) for "xData" instances used for Scenario 2. The field itself specifies that such an "xData" instance SHALL occur at least 1 time and at maximum 3 times within "xListData" of Feature Type "xFeatureType". However, this constraint holds only for Scenario 2 and only if such "xData" are required. In this case, they are required, as the left field

2: M \W

denotes that this data set is mandatory for Scenario 2.

The field

xSlot. (1..)

expresses that the Element "xSlot" SHALL occur at least one time within its "xData", but MAY occur more than one time.

For the expression "<*(1..)>" of Element "xId" please see section 3.1.3.6.

The remaining fields do not have an explicit cardinality indication.

Note: Cardinality expressions are also used within placeholder expressions as defined in section 3.1.3.6. In many cases such placeholder expressions define the number of required or permitted Elements or list items as they explicitly define how many different values for a given Identifier are required or permitted for a given Scenario.

3.1.3.5 Writability and changeability indication

In the same column where the presence indications are denoted, a mark is used to distinguish between writeable, changeable or readable Elements:

- Elements that are marked with "\W" are written by a client and SHALL be writeable at the server according to their presence indications. The client is not obliged to read the according data. Received notifications do not need to be evaluated.
- Elements that are marked with "\C" are changed by a client and SHALL be changeable at the server according to their presence indications. The client is not obliged to read the according data. Received notifications do not need to be evaluated.
- Elements that are marked with "\RW" are read and written by a client and SHALL be writeable and provided by the server according to their presence indications. Received notifications SHALL be evaluated according to their presence indications.
- Elements that are marked with "\RC" are read and changed by a client and SHALL be changeable and provided by the server according to their presence indications. Received notifications SHALL be evaluated according to their presence indications.
- Elements that are not marked are only read by a client and SHALL be provided by the server according to their presence indications. Received notifications SHALL be evaluated according to their presence indications.

"Writeable" means that the Element and its value may be written by a client. This includes the possibility to modify (if the Element is already present), create (if the Element is not present yet), and

delete the Element. The server SHALL adjust its Function according to the received "write" operation (unless the server cannot accept the "write" operation according to section 3.1.3.3).

"Changeable" means that the Element's value may be changed by a client. If the Element is not present at the resource before, it probably **cannot** be created by the client via the "write" operation. In this case the server MAY decline such a message.

Note: "\W" includes "\C" already.

Note: Depending on the resource a client might need to request a proper binding before the server accepts a "write" operation.

3.1.3.6 "Value" placeholders

3.1.3.6.1 Introduction

Specializations may use placeholders to model relations between different Elements or even list items of different Functions. The main purpose is to declare which Identifier values relate to each other. As a Use Case does not prescribe specific values to be used for a given Identifier, a placeholder like "<x1>" can be used in "Value" columns to express the intended relations.

There are two styles to identify a placeholder that can be referenced:

1. <xM>
2. <xM#S>

where

1. "x" is any alphabetical prefix like "m", "t", "ec", "cc", etc.
2. "M" is a (major) number like "1", "2", "15", etc.
3. "S" is a sub-number like "1", "7", "10", etc.

Examples for the first style are "<ec1>", "<z12>". Examples for the second style are "<p4#2>", "<m22#3>". For a given placeholder, only one of the styles can be used.

In addition, there are also styles for placeholders that do not need to be referenced:

1. <*>
2. <*#S>

The second style is only used with so-called cardinality expressions.

3.1.3.6.2 Uniqueness of placeholders

A given placeholder <xM> or <xM#S> represents the same value throughout a given Use Case specification for a given set of its parent Identifier values. This shall be explained in a brief example:

We assume a list item with PRIMARY IDENTIFIER "pld". It also has a SUB IDENTIFIER "sld" with placeholder "<s1>". Then, each occurrence of "<s1>" represents the same value for a given value of

637 pld. This means that "<s1>" of a list item with pld=1 can differ from "<s1>" of a list item with pld=2.
 638 But it also means that "<s1>" represents the same value in all list items with pld=1.

639 Note: Typically, parent Identifiers like "pld" will also be expressed with a placeholder like "<p5>", e.g.
 640 In this case, the uniqueness rule applies for "<p5>" likewise.

641 Note: The uniqueness also applies for placeholders used as FOREIGN IDENTIFIER.

642

643 3.1.3.6.3 Placeholder expressions with cardinalities

644 For some Identifiers, more than one placeholder is needed. Several notations are used for this
 645 purpose, which make use of cardinality expressions. The general notation is as follows:

646 1. <xM#(a..b)>

647 This is equivalent to this explicit definition:

648 At least a and at maximum b placeholders of this list: <xM#1> <xM#2> ... <xM#b>

649 This means that the implementation of a given Use Case (or Scenario) requires a minimum of "a"
 650 distinct values of the respective Identifier. In total, there can be up to "b" distinct values of the
 651 respective Identifier.

652 Additionally, the following notations may occur:

653 2. <xM#(a..)>

654 This is equivalent to "<xM#(a..b)>" with "b" equal to infinity.

655 3. <xM#(..b)>

656 This is equivalent to "<xM#(a..b)>" with "a" equal to zero.

657 "<xM#(a..)>" has only a lower bound of "a" distinct values, but no upper bound. "<xM#(..b)>", on the
 658 other hand, expresses that the Identifier may not be required at all, but it is permitted to have up to
 659 "b" distinct values.

660 Similarly, the cardinality can be used for placeholders that are not referenced, i.e. <*(a..b)> etc.

661 Note: The cardinality does NOT express which of the sub-numbers have to be used! I.e., it does NOT
 662 mean that the Identifiers <xM#1> ... <xM#a> are always used and just those with larger sub-numbers
 663 (<xM#a+1> ... <xM#b>) are optional. If, for instance, a placeholder expression "<xM#(3..5)>" is given,
 664 it may well happen that an implementation makes use of <xM#2>, <xM#4>, and <xM#5> (i.e., it does
 665 NOT use <xM#1>, <xM#3>). Which sub-numbers are used usually depends on other parts of a
 666 Specialization and their references to required placeholders, which is explained in section 3.1.3.6.4.

667

668 3.1.3.6.4 References to placeholders and relations

669 According to the styles for placeholders that can be referenced, an enumeration value "e" can refer
 670 to a particular placeholder:

671 1. e(-><xM>)

672 2. e(-><xM#S>)

673 This denotes that "e" is to be used with "<xM>" or "<xM#S>", resp.

674 Example: A Specialization contains the Elements "mld" and "phase". "mld" is an Identifier with
675 placeholder definition <m2#(1..3)>. "phase" is a string that permits the values "a", "b", and "c" using
676 this expression:

677 "a"(-><m2#1>)

678 "b"(-><m2#2>)

679 "c"(-><m2#3>)

680 This expresses that the enumeration value "a" is to be used with the placeholder <m2#1>, "b" is to
681 be used with <m2#2> and "c" with <m2#3>.

682 Similarly, a placeholder "yN" can refer to a particular placeholder:

683 3. <yN->xM>

684 4. <yN->xM#S>

685 5. <yN#T->xM>

686 6. <yN#T->xM#S>

687 where "T" is a sub-number of "yN".

688 It is also feasible to associate placeholders with cardinalities:

689 7. <yN#(a..b)->xM#(a..b)>

690 denotes that <yN#1> is to be used with <xM#1>, <yN#2> is to be used with <xM#2>, etc.

691 Note: In this case, the placeholder expressions of yN and xM must have the same cardinality.

692 In some cases, there is a need to express that multiple list items with similar values are feasible or
693 required, but only particular combinations of these different data are permitted. The following
694 example shows that several "fData" list items with different "phase" value are required, but that
695 these list items may only express either the "phase" value combination { "a", "b", "c" } or the "phase"
696 value combination { "a", "abc", "neutral" }. The permitted combinations are defined in a note below a
697 table:

Scenario [...]: M/R/O [W][\C]	Element	Value	[High-Level Mapping] Element and value rules
2: M	F. fListData. fData.		
2: M	zld	<z3#(3..5)>	
2: M	phase	"a"(-><z3#1>)	
		"b"(-><z3#2>)	
		"c"(-><z3#3>)	
		"abc"(-><z3#4>)	
		"neutral"(-><z3#5>)	

698 Table 11: Content of an example Specialization

699 Note: One of the following combinations SHALL be used at least: {<z3#1>, <z3#2>, <z3#3>} or
700 {<z3#1>, <z3#4>, <z3#5>}.
701

702 **3.1.3.7 Rules for content of "Value" column**

703 For a given Scenario, the "Value" column may restrict the permitted content of a Function's Element
704 to one or more particular values. This means that Elements with values deviating from the restriction
705 (i.e. from the permitted values) do not belong to the respective Scenario and need to be considered
706 as if the Element is not set. If more than one particular value is permitted for an Element, the values
707 are in a single line each.

708 If a presence indication is set for the value (in an additional column before the value), the following
709 rules SHALL be applied:

- 710 - "M" means that the value SHALL be supported. This means the value needs to be set at a
711 certain point in time (depending on the value rules) or for a certain Element within a list
712 entry.
- 713 - "R" means that the value SHOULD be supported.
- 714 - "O" means that the value MAY be supported.

715 If all possible values of a given mandatory Element are optional or recommended and this Element is
716 used for the purpose of the respective Scenario, one of the values SHALL be set. If all possible values
717 of a given optional or recommended Element are optional or recommended, this Element MAY
718 contain also other values, but then this Element SHALL NOT be considered as part of the respective
719 Scenario.

720 M, R or O may be combined with the suffix "(event)" to express that a supported value only has to be
721 supported during a certain event and hence does not need to be present at all times. If the event is
722 not active another value may be set. In most cases a High-Level requirement reference for the event
723 is given in the rules column.

724 If no presence indication is set for the value, the following rules SHALL be applied:

- 725 - In case of Elements where the server may set or change an Element on its own (see section
726 3.1.3.5):
 - 727 ○ within the tables in the "Server data - Resources" sections:
 - 728 ■ the server SHALL support at least one of the listed values.
 - 729 ○ within the tables in the "Client data - Specializations" sections:
 - 730 ■ the client SHALL support all listed values.
- 731 - In case of Elements that are writeable or changeable (see section 3.1.3.5):
 - 732 ○ within the tables in the "Server data - Resources" sections:
 - 733 ■ the server SHALL support all listed values.
 - 734 ○ within the tables in the "Client data - Specializations" sections:
 - 735 ■ the client SHALL support at least one of the listed values.

736 Depending on the Element, different values may be used during runtime. If this is the case, those
737 rules are described within the value rules.

738 If a value is placed in parenthesis, the corresponding value is a recommendation. The actual value
 739 MAY deviate from this, e.g. "(1024)".

740

741 **3.1.3.8 General information on how to interpret the "Content of Function..." and "Content of** 742 **Specialization..." tables**

743 Within the "Client data - Specializations" sections each Specialization is described in an own sub-
 744 section with the name "Specialization "<name of the Specialization>" (e.g. "Specialization
 745 "Measurement_GridFeedInEnergy"). It contains only one table that includes all Elements needed for
 746 this Specialization. The different Functions are mentioned in a continuous row, highlighted with grey
 747 background colour. This row contains the following parts:

748 <Feature Type>. <Function>.[<list entry instance name>.]

749 The <list entry instance name> is only included if the <Function> is a list-based Function. An example
 750 could be:

751 DeviceConfiguration. deviceConfigurationKeyValueDescriptionListData.
 752 deviceConfigurationKeyValueDescriptionData.

753 In the following rows, only the names of the Elements are stated, without the prefix described above.

754

755 Within the "Server data - Resources" sections each Feature Type is described in an own sub-section
 756 with the name "Feature Type "<name of the Feature Type>" (e.g. "Feature Type "Measurement").
 757 It contains sub-sections for each Function named "Function "<name of the Function>" (e.g.
 758 "Function "measurementListData"). These sections contain one table with all Elements needed for
 759 this resource. The list entries are mentioned in a continuous row, highlighted with grey background
 760 colour. This row contains the following parts:

761 <Feature Type>. <Function>.[<list entry instance name>.]

762 The <list entry instance name> is only included if the <Function> is a list-based Function. An example
 763 could be:

764 Measurement. measurementDescriptionListData. measurementDescriptionData.

765 In the following rows, only the names of the Elements are stated, without the prefix described above.

766

767 For both kinds of tables, the following applies:

768 - Parent Elements are marked with a dot at the end of the name:

769 <parent Element>.

770 E.g.:

771 value.

772 - If there are sub-Elements, they are described in own rows with the name of the parent
 773 Element as prefix, separated by a dot and a blank space:

774 <parent Element>. <sub-Element>
 775 E.g.:
 776 value. number

777

778 3.1.4 Rules for "Feature Types and Functions..." tables

779 3.1.4.1 Presence indications for "Feature Types and Functions..." tables

780 The following presence indications are used:

Abbreviation	Meaning	Link to requirement keywords
M	Mandatory	SHALL
R	Recommended	SHOULD
O	Optional	MAY

781 Table 12: Presence indication of Feature Types and Functions support

782 If at least one Function of a Feature has the presence indication "M", it is mandatory to support the
 783 Feature.

784

785 3.1.4.2 Rules for "Possible operations" column

786 Within the "Feature Types and Functions..." tables the column "Possible operations" state whether
 787 the Function is read- or writeable (as defined in the detailed discovery mechanism, see
 788 [ProtocolSpecification]).

789 If the "partial" concept (also called "restricted function exchange") SHALL be supported, the
 790 following notation is used (separated for read and write access):

791 read (M). partial (M)
 792 write (M). partial (M)

793 If the "partial" concept SHOULD be supported, the following notation is used:

794 read (M). partial (R)
 795 write (M). partial (R)

796 If the "partial" concept MAY be supported, the following notation is used:

797 read (M). partial (O)
 798 write (M). partial (O)

799 The server can decide whether a notification is submitted complete or partial (as described in
 800 [ProtocolSpecification]) if not defined differently within this Use Case Specification.

801

802 3.1.5 "Actor ... overview" diagram rules

803 Within the "Actor [...] overview" diagrams in the "Actors" sub-sections the complete functionality of
 804 this Use Case is provided, including optional Scenarios. Which Scenarios are optional can be found in
 805 Table 1. The Actor MAY have more functionality implemented than needed for this Use Case.

For the following Actor overview example, a brief description of the graphical symbols will be described.

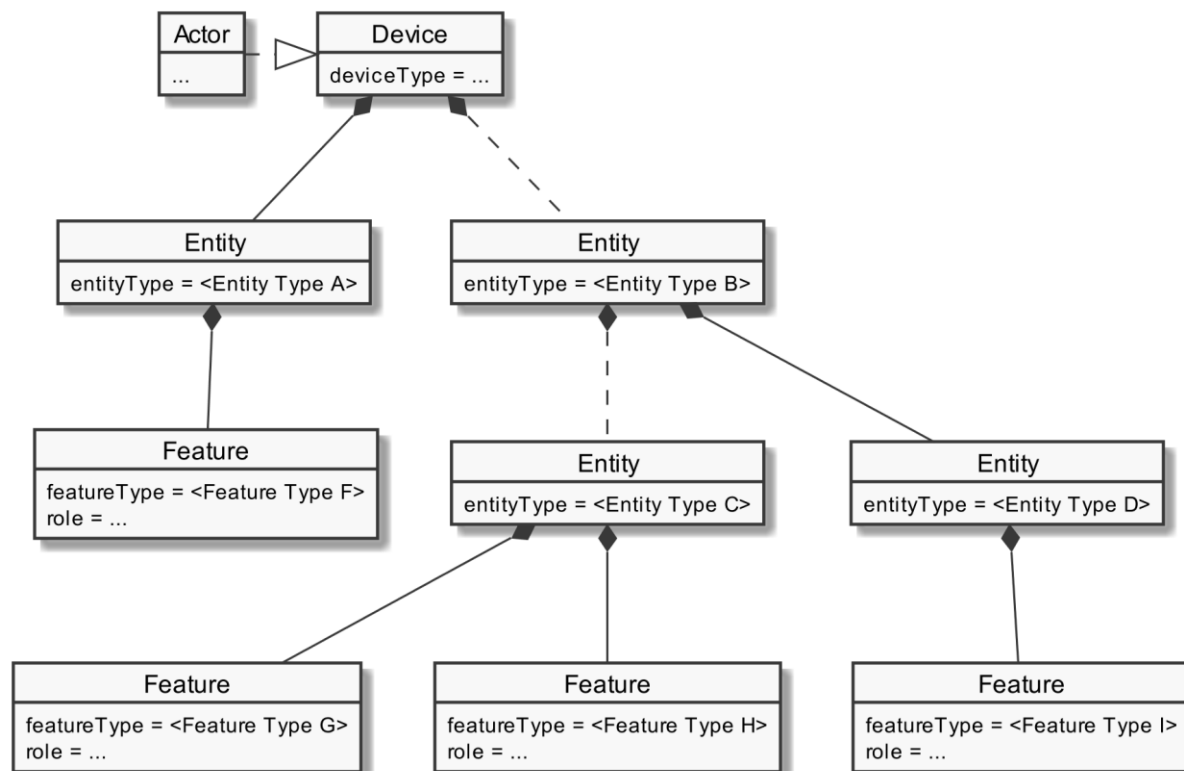


Figure 4: Actor overview example

The solid lines in the figure represent an immediate parent-childhood relation: The Entity with "<Entity Type A>" is a direct child of "Device". The Entity with "<Entity Type D>" is a direct child of the Entity with "<Entity Type B>". All Features are immediate child of the respective Entity.

The dashed lines in the figure express that there MAY be additional Entities between the shown Entities: A vendor's implementation MAY have one or more Entities between "Device" and the Entity with "<Entity Type B>". Likewise, a vendor's implementation MAY have one or more Entities between the Entity with "<Entity Type B>" and the Entity with "<Entity Type C>".

3.1.6 Specializations

Within the "Actors" sub-sections, Specializations are referenced. A Specialization describes a dataset necessary to fulfil the specific requirements of a High-Level Use Case and its Scenarios. Often, data from multiple different Features and Functions are needed to fulfil the requirements. Therefore, a Specialization defines a dataset that may encompass multiple related Functions from one or more different Features.

As different Use Cases sometimes share similar requirements, Specializations are also important from a re-usability perspective. This approach is used to improve consistency across Use Cases and avoid multiple variances of basically the same dataset. This is especially important in the case when an implementation supports multiple Use Cases. E.g. if a power measurement is necessary in two different Use Cases, both Use Cases could define slightly different datasets. In this case, the server as

well as the client functionality would have to implement both variances if both Use Cases are supported. This means, depending on the number of Use Cases, two or more datasets need to be generated, transmitted and stored instead of one. Therefore, common Specializations are defined to enable consolidated implementations. Specializations are detailed per Actor, as distinct Actors may use different parts of a Specialization or even have different permissions to change data. In general, this may also depend on the respective Scenario.

If a Feature server can provide the data of a Specialization, the data does not necessarily always need to be available at the Feature server. There might be situations where the user deactivates a Use Case. There may also be other reasons why Use Case data cannot be provided currently. Therefore, a client always needs to be subscribed (as described in section 3.3.4) on the corresponding dataset to stay updated.

The SPINE resource descriptions given in the "SPINE resources of the Actor" sections are derived from the Specializations given in the Actor's overview diagram.

3.1.7 Order of messages within the sequence diagrams

There are several sequence diagrams in this document describing message flows. The order of the messages SHOULD be kept by the communications partners, but there might be cases where a different order makes sense. The communications partners SHALL be able to handle the Scenario functionalities even if the messages are transmitted in a different order by the other Actor(s). The sequence diagrams can be seen as examples.

3.1.8 Further information and rules

3.1.8.1 Frequently used Element rules for the Resource and Specialization tables

This section serves as a collection of rules frequently used by Resource and Specialization tables of the subsequent sections. Each rule applies only where referenced explicitly in the tables.

Note: The purpose of this collection is just to reduce the size of the tables. As such, no rule has a meaning without a reference indicating the required rule. A reference looks like "See [Measurement value rules]", e.g.

[Measurement value rules]:

SHALL be set if a value is available. Otherwise, the whole list entry SHALL be omitted or the Element *valueState* SHALL be set to "error".

If *valueState* is set to "error", but *value* is set, the content of *value* SHALL be ignored.

If *valueState* is set to "outOfRange", but *value* is set, the content of *value* SHALL be interpreted as being out of range.

If *valueState* is set to "outOfRange", *measurementConstraintsListData.valueRangeMax* is set and *value* is equal or bigger than *valueRangeMax*, *value* SHALL be interpreted as above *valueRangeMax*.

866 If *valueState* is set to "outOfRange", *measurementConstraintsListData.valueRangeMin* is set and
867 *value* is equal or smaller than *valueRangeMin*, *value* SHALL be interpreted as below *valueRangeMin*.

868 If set, *measurementDescriptionListData.measurementType* SHALL be set, too.

869

870 **[Value state rules]:**

871 The Element *valueState* SHALL be set by the server if its content differs from "normal" ([MGCP-003]).
872 This means, if the state of the value is "outOfRange" or "error" this SHALL be denoted in the
873 *valueState* Element. A client SHALL always consider the content of *valueState*, if set. If omitted, a
874 value of "normal" is assumed.

875

876 **[Scaled number rules]:**

877 The sub-Elements "number" and "scale" represent a value according to the following formula:

878 $\text{number} * 10^{\text{scale}}$

879

880 **3.1.8.2 Applied sign convention**

881 For the applied sign convention, please see section 2.6.1.

882

883 **3.1.8.3 Further rules**

884 A server SHOULD NOT add or remove Entities and Features used within this Use Case during runtime
885 in the detailed discovery.

886

887 **3.2 Actors**

888 **3.2.1 Monitoring Appliance**

889 **3.2.1.1 Resource hierarchy**

890 If Use Case discovery is supported (see section 3.1.2) this Actor SHALL be denoted as
891 "MonitoringAppliance" in the Element "nodeManagementUseCaseData.useCaseInformation.actor".

892 The following diagram provides an overview of the Actor Monitoring Appliance resource hierarchy.

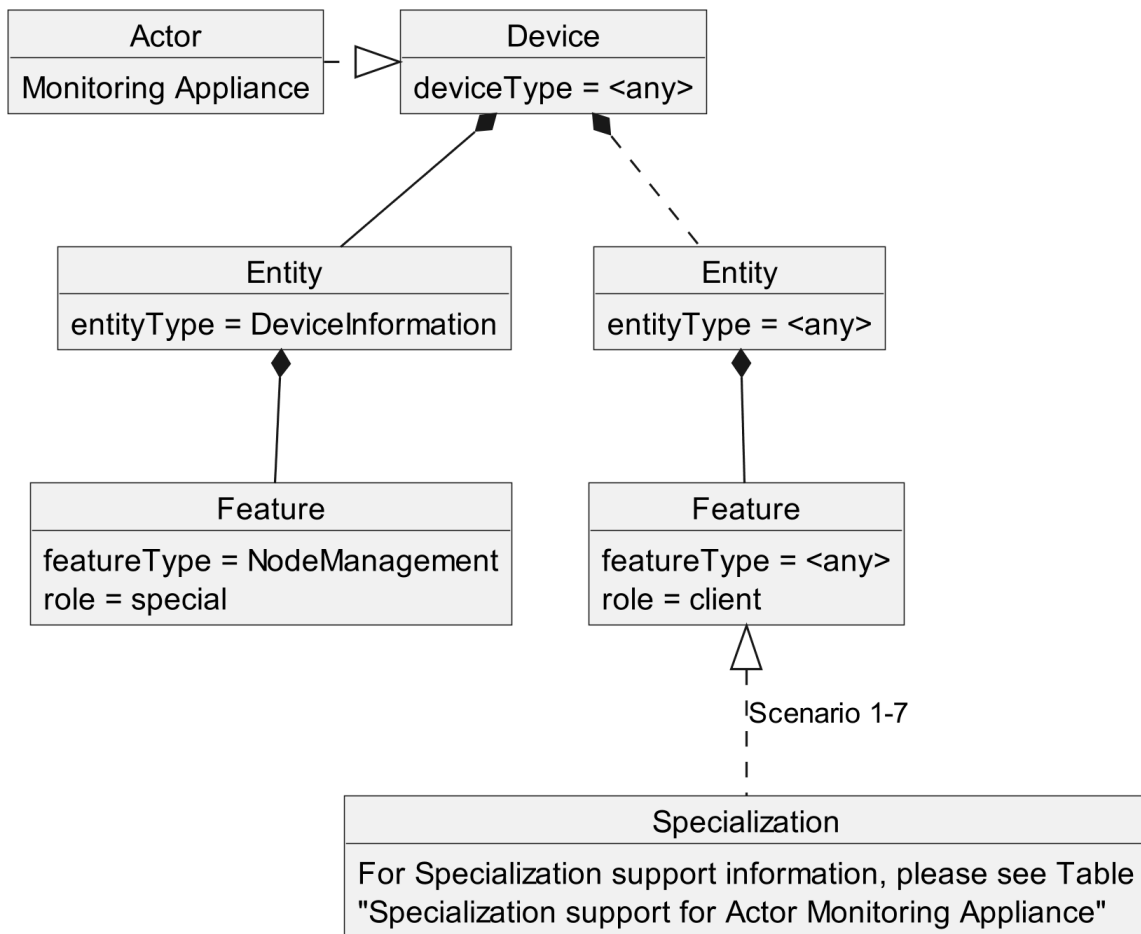


Figure 5: Actor "Monitoring Appliance" overview

The "Actor ... overview" diagram rules" section describes how to interpret the diagram above. See the "Specializations" section for more information regarding the Specializations given in the diagram above.

Note: The entityType "DeviceInformation" with the featureType "NodeManagement" is required by the SPINE protocol and therefore SHALL be supported. Both types are added in the figure for completeness but are not directly linked to the Use Case.

The Use Case specific data follow behind any entityType. The Specializations represent the Scenario specific data that must be supported for each Scenario and are realized through the corresponding featureTypes.

If a Specialization is connected to a Feature with the role "client", the Actor has a client role for this data. This means that the Actor accesses the data set described by the Specialization at a corresponding server Feature. Further details are described in the sub-section "Client data - Specializations".

If a Specialization is connected to a Feature with the role "server", the Actor has the server role for this data. This means that the Actor must provide the corresponding data set of the Specialization as part of its Features. Further details are described in the sub-section "Server data - Resources".

Specialization name	Scenario	Described in table
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DeviceConfiguration_PvCurtailmentLimitFactor	1	Table 14
Measurement_AcPowerTotal	2	Table 15
Measurement_GridFeedInEnergy	3	Table 16
Measurement_GridConsumptionEnergy	4	Table 17
Measurement_AcCurrentPhaseSpecific	5	Table 18
Measurement_AcVoltagePhaseSpecific	6	Table 19
Measurement_AcFrequency	7	Table 20

Table 13: Specialization support for Actor Monitoring Appliance

3.2.1.2 Server data - Resources

As this Actor has only client functionality, no resources are described within this section.

3.2.1.3 Client data - Specializations

3.2.1.3.1 Topic "DeviceConfiguration"

3.2.1.3.1.1 Specialization "DeviceConfiguration_PvCurtailmentLimitFactor"

Scenario {...}: M/R/O [W]/[C]	Element	Value	[High-Level Mapping] Element and value rules
1: M	DeviceConfiguration.deviceConfigurationKeyValueDescriptionListData. deviceConfigurationKeyValueDescriptionData.		
1: M	keyId	<k1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
1: M	keyName	"pvCurtailmentLimitFactor"	The corresponding value SHALL be in the range from 0 to 100.
1: M	valueType	"scaledNumber"	
1: M	unit	"pct"	The unit SHALL be applied to the value of the key.
1: M	DeviceConfiguration.deviceConfigurationKeyValueListData. deviceConfigurationKeyValueData.		
1: M	keyId	<k1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
1: M	value.		[MGCP-011] Exactly one of the child Elements SHALL be set. This SHALL match with the content of <i>valueType</i> Element within the key value description part (see above).
1: M	value. scaledNumber		SHALL be used. The sub-Elements "number" and "scale" represent a value according to the following formula: $\text{number} * 10^{\text{scale}}$
1: M	value. scaledNumber. number		SHALL be used.

1: M	value. scaledNumber. scale		SHALL be interpreted. If absent, a default value of "0" applies.
------	----------------------------------	--	--

Table 14: Content of Specialization "DeviceConfiguration_PvCurtailementLimitFactor" at Actor Monitoring Appliance

3.2.1.3.2 Topic "Measurement"

3.2.1.3.2.1 Specialization "Measurement_AcPowerTotal"

Scenario [{...}]: M/R/O [W][VC]	Element	Value	[High-Level Mapping] Element and value rules
2: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
2: M	measurementId	<m1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M	measurementType	"power"	
2: M	commodityType	"electricity"	
2: M	unit	"W"	
2: M	scopeType	"acPowerTotal"	
2: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		
2: M	measurementId	<m1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: R	valueRangeMin.		SHOULD be used. [Scaled number rules]
2: M	valueRangeMin. number		SHALL be used.
2: M	valueRangeMin. scale		SHALL be interpreted. If absent, a default value of "0" applies.
2: R	valueRangeMax.		SHOULD be used. [Scaled number rules]
2: M	valueRangeMax. number		SHALL be used.
2: M	valueRangeMax. scale		SHALL be interpreted. If absent, a default value of "0" applies.
2: R	valueStepSize.		SHOULD be used. [Scaled number rules]
2: M	valueStepSize. number		SHALL be used.
2: M	valueStepSize. scale		SHALL be interpreted. If absent, a default value of "0" applies.
2: M	Measurement. measurementListData. measurementData.		
2: M	measurementId	<m1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
2: O	timestamp	<t#(1..1)->m1#1>	[MGCP-004] MAY be used. Within this Use Case, only the newest measurement value SHALL be stated. Additional historical values are forbidden.
2: M	value.		[MGCP-001], [MGCP-005], [MGCP-021] [Measurement value rules] [Scaled number rules]

2: M	value. number		SHALL be used.
2: M	value. scale		MAY be used. If absent, a default value of "0" applies.
2: R	valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	
2: M	valueState		[MGCP-003] [Value state rules]
2: M	ElectricalConnection. electricalConnectionDescriptionListData. electricalConnectionDescriptionData.		
2: M	electricalConnectionId	<ec1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
2: M	powerSupplyType	"ac"	
2: M	positiveEnergyDirection	"consume"	
2: M	ElectricalConnection. electricalConnectionParameterDescriptionListData. electricalConnectionParameterDescriptionData.		
2: M	electricalConnectionId	<ec1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
2: M	parameterId	<p1#{1..1}->ec1#1>	SHALL be set as SUB IDENTIFIER.
2: M	measurementId	<m1#{1..1}->p1#1>	SHALL be set as FOREIGN IDENTIFIER.
2: M	voltageType	"ac"	
2: O	acMeasuredPhases	"abc" "ab" "bc" "ac" "a" "b" "c" (->p1#1)	If the Monitored Unit is connected to less than three phases, one of the other combinations like "a" or "ab" are allowed instead of "abc". The values "a", "b", and "c" are permitted if and only if only one phase is connected to the Monitored Unit.
2: O	acMeasuredInReference To	"neutral"	
2: M	acMeasurementType	"real"	
2: O	acMeasurementVariant	"rms"	

Table 15: Content of Specialization "Measurement_AcPowerTotal" at Actor Monitoring Appliance

3.2.1.3.2.2 Specialization "Measurement_GridFeedInEnergy"

Scenario [{...}]: M/R/O [W][C]	Element	Value	[High-Level Mapping] Element and value rules
3: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
3: M	measurementId	<m2#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
3: M	measurementType	"energy"	
3: M	commodityType	"electricity"	
3: M	unit	"Wh"	
3: M	scopeType	"gridFeedIn"	
3: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		
3: M	measurementId	<m2#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
3: R	valueRangeMin.		SHOULD be used.

			[Scaled number rules]
3: M	valueRangeMin. number		SHALL be used.
3: M	valueRangeMin. scale		SHALL be interpreted. If absent, a default value of "0" applies.
3: R	valueRangeMax.		SHOULD be used. [Scaled number rules]
3: M	valueRangeMax. number		SHALL be used.
3: M	valueRangeMax. scale		SHALL be interpreted. If absent, a default value of "0" applies.
3: R	valueStepSize.		SHOULD be used. [Scaled number rules]
3: M	valueStepSize. number		SHALL be used.
3: M	valueStepSize. scale		SHALL be interpreted. If absent, a default value of "0" applies.
3: M	Measurement. measurementListData. measurementData.		
3: M	measurementId	<m2#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
3: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
3: O	timestamp	<t#{1..1}->m2#1>	[MGCP-006] MAY be used. Within this Use Case, only the newest measurement value SHALL be stated. Additional historical values are forbidden.
3: M	value.		[MGCP-001], [MGCP-005], [MGCP-007], [MGCP-031], [Measurement value rules] [Scaled number rules]
3: M	value. number		SHALL be used.
3: M	value. scale		MAY be used. If absent, a default value of "0" applies.
3: M	valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	
3: M	valueState		[MGCP-003] [Value state rules]
3: M	ElectricalConnection. electricalConnectionDescriptionListData. electricalConnectionDescriptionData.		
3: M	electricalConnectionId	<ec1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
3: M	powerSupplyType	"ac"	
3: M	positiveEnergyDirection	"consume"	
3: M	ElectricalConnection. electricalConnectionParameterDescriptionListData. electricalConnectionParameterDescriptionData.		
3: M	electricalConnectionId	<ec1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
3: M	parameterId	<p2#{1..1}->ec1#1>	SHALL be set as SUB IDENTIFIER.
3: M	measurementId	<m2#{1..1}->p2#1>	SHALL be set as FOREIGN IDENTIFIER.
3: M	voltageType	"ac"	
3: M	acMeasurementType	"real"	

Table 16: Content of Specialization "Measurement_GridFeedInEnergy" at Actor Monitoring Appliance

3.2.1.3.2.3 Specialization "Measurement_GridConsumptionEnergy"

Scenario [...]: M/R/O [W][C]	Element	Value	[High-Level Mapping] Element and value rules
4: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
4: M	measurementId	<m3#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
4: M	measurementType	"energy"	
4: M	commodityType	"electricity"	
4: M	unit	"Wh"	
4: M	scopeType	"gridConsumption"	
4: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		
4: M	measurementId	<m3#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
4: R	valueRangeMin.		SHOULD be used. [Scaled number rules]
4: M	valueRangeMin. number		SHALL be used.
4: M	valueRangeMin. scale		SHALL be interpreted. If absent, a default value of "0" applies.
4: R	valueRangeMax.		SHOULD be used. [Scaled number rules]
4: M	valueRangeMax. number		SHALL be used.
4: M	valueRangeMax. scale		SHALL be interpreted. If absent, a default value of "0" applies.
4: R	valueStepSize.		SHOULD be used. [Scaled number rules]
4: M	valueStepSize. number		SHALL be used.
4: M	valueStepSize. scale		SHALL be interpreted. If absent, a default value of "0" applies.
4: M	Measurement. measurementListData. measurementData.		
4: M	measurementId	<m3#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
4: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
4: O	timestamp	<t#(1..1)->m3#1>	[MGCP-006] MAY be used. Within this Use Case, only the newest measurement value SHALL be stated. Additional historical values are forbidden.

4: M	value.		[MGCP-001], [MGCP-005], [MGCP-008], [MGCP-041] [Measurement value rules] [Scaled number rules]
4: M	value. number		SHALL be used.
4: M	value. scale		MAY be used. If absent, a default value of "0" applies.
4: M	valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	
4: M	valueState		[MGCP-003] [Value state rules]
4: M	ElectricalConnection. electricalConnectionDescriptionListData. electricalConnectionDescriptionData.		
4: M	electricalConnectionId	<ec1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
4: M	powerSupplyType	"ac"	
4: M	positiveEnergyDirection	"consume"	
4: M	ElectricalConnection. electricalConnectionParameterDescriptionListData. electricalConnectionParameterDescriptionData.		
4: M	electricalConnectionId	<ec1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
4: M	parameterId	<p3#{1..1}->ec1#1>	SHALL be set as SUB IDENTIFIER.
4: M	measurementId	<m3#{1..1}->p3#1>	SHALL be set as FOREIGN IDENTIFIER.
4: M	voltageType	"ac"	
4: M	acMeasurementType	"real"	

Table 17: Content of Specialization "Measurement_GridConsumptionEnergy" at Actor Monitoring Appliance

3.2.1.3.2.4 Specialization "Measurement_AcCurrentPhaseSpecific"

Scenario [...]: M/R/O [W][C]	Element	Value	[High-Level Mapping] Element and value rules
5: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
5: M	measurementId	<m4#{1..3}>	SHALL be set as PRIMARY IDENTIFIER.
5: M	measurementType	"current"	
5: M	commodityType	"electricity"	
5: M	unit	"A"	
5: M	scopeType	"acCurrent"	
5: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		
5: M	measurementId	<m4#{1..3}>	SHALL be set as PRIMARY IDENTIFIER.
5: R	valueRangeMin.		SHOULD be used. [Scaled number rules]

5: M	valueRangeMin. number		SHALL be used.
5: M	valueRangeMin. scale		SHALL be interpreted. If absent, a default value of "0" applies.
5: R	valueRangeMax.		SHOULD be used. [Scaled number rules]
5: M	valueRangeMax. number		SHALL be used.
5: M	valueRangeMax. scale		SHALL be interpreted. If absent, a default value of "0" applies.
5: R	valueStepSize.		SHOULD be used. [Scaled number rules]
5: M	valueStepSize. number		SHALL be used.
5: M	valueStepSize. scale		SHALL be interpreted. If absent, a default value of "0" applies.
5: M	Measurement. measurementListData. measurementData.		
5: M	measurementId	<m4#(1..3)>	SHALL be set as PRIMARY IDENTIFIER.
5: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
5: O	timestamp	<t#(1..1)->m4#(1..3)>	[MGCP-004] MAY be used. Within this Use Case, only the newest measurement value SHALL be stated. Additional historical values are forbidden.
5: M	value.		[MGCP-001], [MGCP-051] [Measurement value rules] See [Scaled number rules].
5: M	value. number		SHALL be used.
5: M	value. scale		MAY be used. If absent, a default value of "0" applies.
5: R	valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	
5: M	valueState		[MGCP-003] [Value state rules]
5: M	ElectricalConnection. electricalConnectionDescriptionListData. electricalConnectionDescriptionData.		
5: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
5: M	powerSupplyType	"ac"	
5: M	positiveEnergyDirection	"consume"	
5: M	ElectricalConnection. electricalConnectionParameterDescriptionListData. electricalConnectionParameterDescriptionData.		
5: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
5: M	parameterId	<p4#(1..3)->ec1#1>	SHALL be set as SUB IDENTIFIER.
5: M	measurementId	<m4#(1..3)->p4#(1..3)>	SHALL be set as FOREIGN IDENTIFIER.

5: M	voltageType	"ac"	
5: M	acMeasuredPhases	"a" (-><p4#1>)* ¹	[MGCP-051/1]
		"b" (-><p4#2>)* ¹	[MGCP-051/2]
		"c" (-><p4#3>)* ¹	[MGCP-051/3]
5: M	acMeasurementType	"real"	[MGCP-050/1]
5: M	acMeasurementVariant	"rms"	[MGCP-050/2]

Table 18: Content of Specialization "Measurement_AcCurrentPhaseSpecific" at Actor Monitoring Appliance

*¹: Each permitted value of the Element "acMeasuredPhases" SHALL NOT be used for more than one value of Element "parameterId".

3.2.1.3.2.5 Specialization "Measurement_AcVoltagePhaseSpecific"

Scenario {...}: M/R/O [W]/[C]	Element	Value	[High-Level Mapping] Element and value rules
6: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
6: M	measurementId	<m5#(1..6)>	SHALL be set as PRIMARY IDENTIFIER.
6: M	measurementType	"voltage"	
6: M	commodityType	"electricity"	
6: M	unit	"V"	
6: M	scopeType	"acVoltage"	
6: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		
6: M	measurementId	<m5#(1..6)>	SHALL be set as PRIMARY IDENTIFIER.
6: R	valueRangeMin.		SHOULD be used. [Scaled number rules]
6: M	valueRangeMin. number		SHALL be used.
6: M	valueRangeMin. scale		SHALL be interpreted. If absent, a default value of "0" applies.
6: R	valueRangeMax.		SHOULD be used. [Scaled number rules]
6: M	valueRangeMax. number		SHALL be used.
6: M	valueRangeMax. scale		SHALL be interpreted. If absent, a default value of "0" applies.
6: R	valueStepSize.		SHOULD be used. [Scaled number rules]
6: M	valueStepSize. number		SHALL be used.
6: M	valueStepSize. scale		SHALL be interpreted. If absent, a default value of "0" applies.
6: M	Measurement. measurementListData. measurementData.		
6: M	measurementId	<m5#(1..6)>	SHALL be set as PRIMARY IDENTIFIER.

6: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
6: O	timestamp	<t#{1..1}->m5#{1..6}>	MAY be used. Only the newest measurement value SHALL be stated. Additional historical values are forbidden.
6: M	value.		[MGCP-002], [MGCP-061] [Measurement value rules] [Scaled number rules]
6: M	value. number		SHALL be used.
6: M	value. scale		SHALL be interpreted. If absent, a default value of "0" applies.
6: R	valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	
6: M	valueState		[MGCP-003] [Value state rules]
6: M	ElectricalConnection. electricalConnectionDescriptionListData. electricalConnectionDescriptionData.		
6: M	electricalConnectionId	<ec1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
6: M	powerSupplyType	"ac"	
6: M	positiveEnergyDirection	"consume"	
6: M	ElectricalConnection. electricalConnectionParameterDescriptionListData. electricalConnectionParameterDescriptionData.		
6: M	electricalConnectionId	<ec1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
6: M	parameterId	<p5#{1..6}->ec1#1>	SHALL be set as SUB IDENTIFIER.
6: M	measurementId	<m5#{1..6}->p5#{1..6}>	SHALL be set as FOREIGN IDENTIFIER.
6: M	voltageType	"ac"	
6: M	acMeasuredPhases	"a" (-><p5#1>)	[MGCP-061/1]
		"a" (-><p5#4>)	[MGCP-061/4]
		"b" (-><p5#2>)	[MGCP-061/2]
		"b" (-><p5#5>)	[MGCP-061/5]
		"c" (-><p5#3>)	[MGCP-061/3]
		"c" (-><p5#6>)	[MGCP-061/6]
6: M	acMeasuredInReferenceTo	"a" (-><p5#6>)	[MGCP-061/6]
		"b" (-><p5#4>)	[MGCP-061/4]
		"c" (-><p5#5>)	[MGCP-061/5]
		"neutral" (-><p5#1>)	[MGCP-061/1]
		"neutral" (-><p5#2>)	[MGCP-061/2]
		"neutral" (-><p5#3>)	[MGCP-061/3]
6: M	acMeasurementType	"apparent"	
6: M	acMeasurementVariant	"rms"	

Table 19: Content of Specialization "Measurement_AcVoltagePhaseSpecific" at Actor Monitoring Appliance

Note: The Specialization permits up to six phase measurements: Measurement of phase "a" to "neutral", phase "a" to phase "b", phase "b" to "neutral", etc.

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941 3.2.1.3.2.6 Specialization "Measurement_AcFrequency"

Scenario [...]: M/R/O [W][C]	Element	Value	[High-Level Mapping] Element and value rules
7: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
7: M	measurementId	<m6#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
7: M	measurementType	"frequency"	
7: M	commodityType	"electricity"	
7: M	unit	"Hz"	
7: M	scopeType	"acFrequency"	
7: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		
7: M	measurementId	<m6#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
7: R	valueRangeMin.		SHOULD be used. [Scaled number rules]
7: M	valueRangeMin. number		SHALL be used.
7: M	valueRangeMin. scale		SHALL be interpreted. If absent, a default value of "0" applies.
7: R	valueRangeMax.		SHOULD be used. [Scaled number rules]
7: M	valueRangeMax. number		SHALL be used.
7: M	valueRangeMax. scale		SHALL be interpreted. If absent, a default value of "0" applies.
7: R	valueStepSize.		SHOULD be used. [Scaled number rules]
7: M	valueStepSize. number		SHALL be used.
7: M	valueStepSize. scale		SHALL be interpreted. If absent, a default value of "0" applies.
7: M	Measurement. measurementListData. measurementData.		
7: M	measurementId	<m6#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
7: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
7: O	timestamp	<t#{1..1}->m6#{1..1}>	MAY be used. Only the newest measurement value SHALL be stated. Additional historical values are forbidden.
7: M	value.		[MGCP-071] [Measurement value rules] [Scaled number rules]
7: M	value. number		SHALL be used.

7: M	value. scale		SHALL be interpreted. If absent, a default value of "0" applies.
7: R	valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	
7: M	valueState		[MGCP-003] [Value state rules]
7: M	ElectricalConnection. electricalConnectionDescriptionListData. electricalConnectionDescriptionData.		
7: M	electricalConnectionId	<ec1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
7: M	powerSupplyType	"ac"	
7: M	positiveEnergyDirection	"consume"	
7: M	ElectricalConnection. electricalConnectionParameterDescriptionListData. electricalConnectionParameterDescriptionData.		
7: M	electricalConnectionId	<ec1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
7: M	parameterId	<p6#{1..1}->ec1#1>	SHALL be set as SUB IDENTIFIER.
7: M	measurementId	<m6#{1..1}->p6#1>	SHALL be set as FOREIGN IDENTIFIER.
7: M	voltageType	"ac"	

Table 20: Content of Specialization "Measurement_AcFrequency" at Actor Monitoring Appliance

3.2.2 Grid Connection Point

3.2.2.1 Resource hierarchy

If Use Case discovery is supported (see section 3.1.2) this Actor SHALL be denoted as "GridConnectionPoint" in the Element "nodeManagementUseCaseData. useCaseInformation. actor".

The following diagram provides an overview of the Actor Grid Connection Point resource hierarchy.

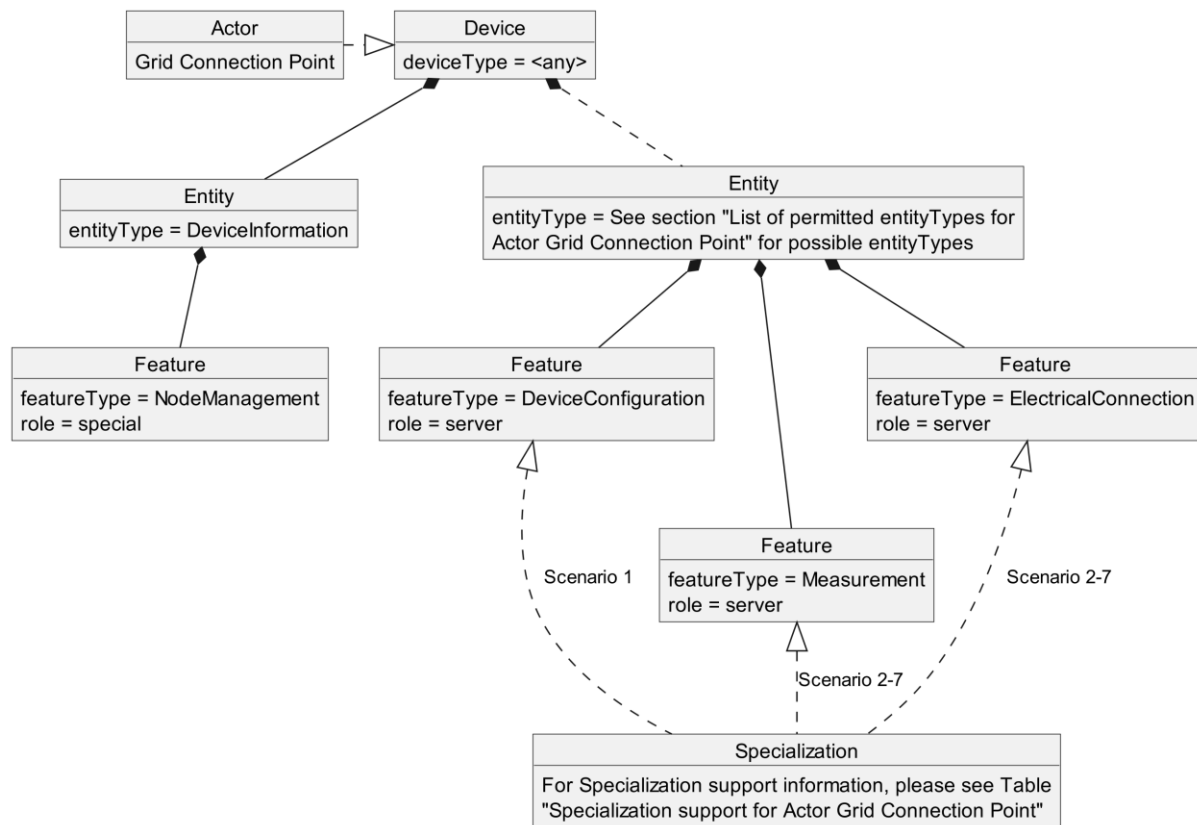


Figure 6: Actor "Grid Connection Point" overview

The "Actor ... overview" diagram rules" section describes how to interpret the diagram above. See the "Specializations" section for more information regarding the Specializations given in the diagram above.

Note: The entityType "DeviceInformation" with the featureType "NodeManagement" is required by the SPINE protocol and therefore SHALL be supported. Both types are added in the figure for completeness but are not directly linked to the Use Case.

The Use Case specific data follow behind one of the entityTypes listed in section 3.2.2.1.1. The Specializations represent the Scenario specific data that must be supported for each Scenario and are realized through the corresponding featureTypes.

If a Specialization is connected to a Feature with the role "client", the Actor has a client role for this data. This means that the Actor accesses the data set described by the Specialization at a corresponding server Feature. Further details are described in the sub-section "Client data - Specializations".

If a Specialization is connected to a Feature with the role "server", the Actor has the server role for this data. This means that the Actor must provide the corresponding data set of the Specialization as part of its Features. Further details are described in the sub-section "Server data - Resources".

Specialization name	Scenario	Used Feature...	...in tables
DeviceConfiguration_PvCurtailementLimitFactor	1	DeviceConfiguration	Table 23
			Table 24
Measurement_AcPowerTotal	2	ElectricalConnection	Table 28

			Table 29
		Measurement	Table 25
			Table 26
			Table 27
Measurement_GridFeedInEnergy	3	ElectricalConnection	Table 28
			Table 29
		Measurement	Table 25
			Table 26
Measurement_GridConsumptionEnergy	4	ElectricalConnection	Table 28
			Table 29
		Measurement	Table 25
			Table 26
Measurement_AcCurrentPhaseSpecific	5	ElectricalConnection	Table 28
			Table 29
		Measurement	Table 25
			Table 26
Measurement_AcVoltagePhaseSpecific	6	ElectricalConnection	Table 28
			Table 29
		Measurement	Table 25
			Table 26
Measurement_AcFrequency	7	ElectricalConnection	Table 28
			Table 29
		Measurement	Table 25
			Table 26
			Table 27

Table 21: Specialization support for Actor Grid Connection Point

3.2.2.1.1 List of permitted entityTypes for Actor Grid Connection Point

In this version of this Use Case, only the following entityTypes are permitted for the Actor Grid Connection Point:

- CEM
- GridConnectionPointOfPremises

This may change at a later version and be extended to further entityTypes.

Note: If located on the entityType "CEM", the Actor Grid Connection Point placed there is a surrogate for the actual Grid Connection Point. Hence, the CEM copies the values from the actual Grid Connection Point device and provides them to further appliances interested in the MGCP values.

3.2.2.2 Server data - Resources

3.2.2.2.1 Overview

Behind one of the entityTypes listed in section 3.2.2.1.1, the Actor Grid Connection Point SHALL offer the Feature Types and Functions given in the table below.

Feature Type	Scenario: M/R/O	Function	Possible operations
DeviceConfiguration	1: M	deviceConfigurationKeyValueDescriptionListData	read (M). partial (R)
	1: M	deviceConfigurationKeyValueListData	read (M). partial (R)
Measurement	2: M 3: M 4: M 5: M 6: M 7: M	measurementDescriptionListData	read (M). partial (R)
	2: R 3: R 4: R 5: R 6: R 7: R	measurementConstraintsListData	read (M). partial (R)
	2: M 3: M 4: M 5: M 6: M 7: M	measurementListData	read (M). partial (R)
ElectricalConnection	2: M 3: M 4: M 5: M 6: M 7: M	electricalConnectionDescriptionListData	read (M). partial (R)
	2: M 3: M 4: M 5: M 6: M 7: M	electricalConnectionParameterDescriptionListData	read (M). partial (R)

Table 22: Feature Types and Functions used within this Use Case by the Actor Grid Connection Point

For each of these Feature Types, the following rule applies: There SHALL be at maximum one Feature with the Feature Type in the Entity.

Note: As a consequence of the previous rule, an implementation may need to have Feature data from different Scenarios/Specializations or even Use Cases in a given Feature.

The Scenario number shows in which Scenarios the Grid Connection Point acts as server and which Feature Types and Functions are relevant in each Scenario.

A detailed definition of the Elements and values that shall be supported in each Function is given in the following sub-sections.

Note: If in the table above "partial" read is not mentioned or is only optional, it still might be mandatory to support partial notifications. The details of "partial" support are described within the Scenario sections.

Note: The presence indications stated above are meant relative to the ones of the according Scenario stated in Table 1. I.e., if a Scenario is optional ("O") and a Feature Type is mandatory ("M"), the Feature Type need only be supported if the Scenario is supported, too.

Note: Further Features MAY be implemented on the same Entities; also, further Functions MAY be implemented in the used Entities.

3.2.2.2.2 Feature Type "DeviceConfiguration"

3.2.2.2.2.1 Function "deviceConfigurationKeyValueDescriptionListData"

Scenario [{...}]: M/R/O [W][C]	Element	Value	[High-Level Mapping] Element and value rules
1: M	DeviceConfiguration. deviceConfigurationKeyValueDescriptionListData. deviceConfigurationKeyValueDescriptionData.		
1: M	keyId	<k1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
1: M	keyName	"pvCurtaimentLimitFactor"	The corresponding value SHALL be in the range from 0 to 100.
1: M	valueType	"scaledNumber"	
1: M	unit	"pct"	The unit SHALL be applied to the value of the key.

Table 23: Content of Function "deviceConfigurationKeyValueDescriptionListData" at Actor Grid Connection Point

3.2.2.2.2.2 Function "deviceConfigurationKeyValueListData"

Scenario [{...}]: M/R/O [W][C]	Element	Value	[High-Level mapping] Element and value rules
1: M	DeviceConfiguration. deviceConfigurationKeyValueListData. deviceConfigurationKeyValueData.		
1: M	keyId	<k1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
1: M	value.		[MGCP-011] Exactly one of the child Elements SHALL be set. This SHALL match with the content

			of <i>valueType</i> Element within the key value description part (see above).
1: M	value. scaledNumber		SHALL be used. [Scaled number rules]
1: M	value. scaledNumber. number		SHALL be used.
1: O	value. scaledNumber. scale		MAY be used. If absent, a default value of "0" applies.

Table 24: Content of Function "deviceConfigurationKeyValueListData" at Actor Grid Connection Point

3.2.2.2.3 Feature Type "Measurement"

3.2.2.2.3.1 Function "measurementDescriptionListData"

Scenario [...]: M/R/O [W][C]	Element	Value	[High-Level Mapping] Element and value rules
2: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
2: M	measurementId	<m1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
2: M	measurementType	"power"	
2: M	commodityType	"electricity"	
2: M	unit	"W"	
2: M	scopeType	"acPowerTotal"	
3: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
3: M	measurementId	<m2#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
3: M	measurementType	"energy"	
3: M	commodityType	"electricity"	
3: M	unit	"Wh"	
3: M	scopeType	"gridFeedIn"	
4: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
4: M	measurementId	<m3#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
4: M	measurementType	"energy"	
4: M	commodityType	"electricity"	
4: M	unit	"Wh"	
4: M	scopeType	"gridConsumption"	
5: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
5: M	measurementId	<m4#{1..3}>	SHALL be set as PRIMARY IDENTIFIER.
5: M	measurementType	"current"	
5: M	commodityType	"electricity"	
5: M	unit	"A"	
5: M	scopeType	"acCurrent"	
6: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
6: M	measurementId	<m5#{1..6}>	SHALL be set as PRIMARY IDENTIFIER.
6: M	measurementType	"voltage"	
6: M	commodityType	"electricity"	

6: M	unit	"V"	
6: M	scopeType	"acVoltage"	
7: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
7: M	measurementId	<m6#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
7: M	measurementType	"frequency"	
7: M	commodityType	"electricity"	
7: M	unit	"Hz"	
7: M	scopeType	"acFrequency"	

Table 25: Content of Function "measurementDescriptionListData" at Actor Grid Connection Point

3.2.2.2.3.2 Function "measurementConstraintsListData"

Scenario [...]: M/R/O [W][C]	Element	Value	[High-Level Mapping] Element and value rules
2: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		
2: M	measurementId	<m1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
2: R	valueRangeMin.		SHOULD be used. [Scaled number rules]
2: M	valueRangeMin. number		SHALL be used.
2: O	valueRangeMin. scale		MAY be used. If absent, a default value of "0" applies.
2: R	valueRangeMax.		SHOULD be used. [Scaled number rules]
2: M	valueRangeMax. number		SHALL be used.
2: O	valueRangeMax. scale		MAY be used. If absent, a default value of "0" applies.
2: R	valueStepSize.		SHOULD be used. [Scaled number rules]
2: M	valueStepSize. number		SHALL be used.
2: O	valueStepSize. scale		MAY be used. If absent, a default value of "0" applies.
3: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		
3: M	measurementId	<m2#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
3: O	valueRangeMin.		SHOULD be used. [Scaled number rules]
3: M	valueRangeMin. number		SHALL be used.
3: O	valueRangeMin. scale		MAY be used. If absent, a default value of "0" applies.
3: O	valueRangeMax.		SHOULD be used. [Scaled number rules]
3: M	valueRangeMax. number		SHALL be used.

3: O	valueRangeMax. scale		MAY be used. If absent, a default value of "0" applies.
3: R	valueStepSize.		SHOULD be used. [Scaled number rules]
3: M	valueStepSize. number		SHALL be used.
3: O	valueStepSize. scale		MAY be used. If absent, a default value of "0" applies.
4: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		
4: M	measurementId	<m3#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
4: O	valueRangeMin.		SHOULD be used. [Scaled number rules]
4: M	valueRangeMin. number		SHALL be used.
4: O	valueRangeMin. scale		MAY be used. If absent, a default value of "0" applies.
4: O	valueRangeMax.		SHOULD be used. [Scaled number rules]
4: M	valueRangeMax. number		SHALL be used.
4: O	valueRangeMax. scale		MAY be used. If absent, a default value of "0" applies.
4: R	valueStepSize.		SHOULD be used. [Scaled number rules]
4: M	valueStepSize. number		SHALL be used.
4: O	valueStepSize. scale		MAY be used. If absent, a default value of "0" applies.
5: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		
5: M	measurementId	<m4#{1..3}>	SHALL be set as PRIMARY IDENTIFIER.
5: R	valueRangeMin.		SHOULD be used. [Scaled number rules]
5: M	valueRangeMin. number		SHALL be used.
5: O	valueRangeMin. scale		MAY be used. If absent, a default value of "0" applies.
5: R	valueRangeMax.		SHOULD be used. [Scaled number rules]
5: M	valueRangeMax. number		SHALL be used.
5: O	valueRangeMax. scale		MAY be used. If absent, a default value of "0" applies.
5: R	valueStepSize.		SHOULD be used. [Scaled number rules]
5: M	valueStepSize. number		SHALL be used.
5: O	valueStepSize. scale		MAY be used. If absent, a default value of "0" applies.
6: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		
6: M	measurementId	<m5#{1..6}>	SHALL be set as PRIMARY IDENTIFIER.
6: R	valueRangeMin.		SHOULD be used. [Scaled number rules]

6: M	valueRangeMin. number		SHALL be used.
6: O	valueRangeMin. scale		MAY be used. If absent, a default value of "0" applies.
6: R	valueRangeMax.		SHOULD be used. [Scaled number rules]
6: M	valueRangeMax. number		SHALL be used.
6: O	valueRangeMax. scale		MAY be used. If absent, a default value of "0" applies.
6: R	valueStepSize.		SHOULD be used. [Scaled number rules]
6: M	valueStepSize. number		SHALL be used.
6: O	valueStepSize. scale		MAY be used. If absent, a default value of "0" applies.
7: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		
7: M	measurementId	<m6#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
7: R	valueRangeMin.		SHOULD be used. [Scaled number rules]
7: M	valueRangeMin. number		SHALL be used.
7: O	valueRangeMin. scale		SHALL be interpreted. If absent, a default value of "0" applies.
7: R	valueRangeMax.		SHOULD be used. [Scaled number rules]
7: M	valueRangeMax. number		SHALL be used.
7: O	valueRangeMax. scale		SHALL be interpreted. If absent, a default value of "0" applies.
7: R	valueStepSize.		SHOULD be used. [Scaled number rules]
7: M	valueStepSize. number		SHALL be used.
7: O	valueStepSize. scale		SHALL be interpreted. If absent, a default value of "0" applies.

1013 Table 26: Content of Function "measurementConstraintsListData" at Actor Grid Connection Point

1014

1015 3.2.2.2.3.3 Function "measurementListData"

Scenario [{...}]: M/R/O [W]\C]	Element	Value	[High-Level Mapping] Element and value rules
2: M	Measurement. measurementListData. measurementData.		
2: M	measurementId	<m1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
2: O	timestamp	<t#(1..1)->m1#1>	[MGCP-004]

			MAY be used. Within this Use Case, only the newest measurement value SHALL be stated. Additional historical values are forbidden.
2: M	value.		[MGCP-001], [MGCP-005], [MGCP-021] [Measurement value rules] [Scaled number rules]
2: M	value. number		SHALL be used.
2: O	value. scale		MAY be used. If absent, a default value of "0" applies.
2: M	valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	
2: R	valueState		[MGCP-003] [Value state rules]
3: M	Measurement. measurementListData. measurementData.		
3: M	measurementId	<m2#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
3: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
3: O	timestamp	<t#{1..1}->m2#1>	[MGCP-006] MAY be used. Within this Use Case, only the newest measurement value SHALL be stated. Additional historical values are forbidden.
3: M	value.		[MGCP-001], [MGCP-005], [MGCP-007], [MGCP-031] [Measurement value rules] [Scaled number rules]
3: M	value. number		SHALL be used.
3: O	value. scale		MAY be used. If absent, a default value of "0" applies.
3: M	valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	
3: R	valueState		[MGCP-003] [Value state rules]
4: M	Measurement. measurementListData. measurementData.		
4: M	measurementId	<m3#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
4: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
4: O	timestamp	<t#{1..1}->m3#1>	[MGCP-006] MAY be used. Within this Use Case, only the newest measurement value SHALL be stated. Additional historical values are forbidden.
4: M	value.		[MGCP-001], [MGCP-005], [MGCP-008], [MGCP-041] [Measurement value rules] [Scaled number rules]
4: M	value. number		SHALL be used.
4: O	value. scale		MAY be used. If absent, a default value of "0" applies.
4: M	valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	
4: R	valueState		[MGCP-003]

			[Value state rules]
5: M	Measurement. measurementListData. measurementData.		
5: M	measurementId	<m4#(1..3)>	SHALL be set as PRIMARY IDENTIFIER.
5: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
5: O	timestamp	<t#(1..1)->m4#1>	[MGCP-004] MAY be used. Within this Use Case, only the newest measurement value SHALL be stated. Additional historical values are forbidden.
5: M	value.		[MGCP-001], [MGCP-051] [Measurement value rules] [Scaled number rules]
5: M	value. number		SHALL be used.
5: O	value. scale		MAY be used. If absent, a default value of "0" applies.
5: M	valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	
5: R	valueState		[MGCP-003] [Value state rules]
6: M	Measurement. measurementListData. measurementData.		
6: M	measurementId	<m5#(1..6)>	SHALL be set as PRIMARY IDENTIFIER.
6: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
6: O	timestamp	<t#(1..1)->m5#(1..6)>	MAY be used. Only the newest measurement value SHALL be stated. Additional historical values are forbidden.
6: M	value.		[MGCP-002], [MGCP-061] [Measurement value rules] [Scaled number rules]
6: M	value. number		SHALL be used.
6: O	value. scale		MAY be used. If absent, a default value of "0" applies.
6: M	valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	
6: R	valueState		[MGCP-003] [Value state rules]
7: M	Measurement. measurementListData. measurementData.		
7: M	measurementId	<m6#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
7: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
7: O	timestamp	<t#(1..1)->m6#1>	MAY be used. Only the newest measurement value SHALL be stated. Additional historical values are forbidden.
7: M	value.		[MGCP-071] [Measurement value rules] [Scaled number rules]
7: M	value. number		SHALL be used.
7: O	value. scale		SHALL be interpreted. If absent, a default value of "0" applies.
7: M	valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	

7: R	valueState		[MGCP-003] [Value state rules]
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Table 27: Content of Function "measurementListData" at Actor Grid Connection Point

3.2.2.2.4 Feature Type "ElectricalConnection"

3.2.2.2.4.1 Function "electricalConnectionDescriptionListData"

Scenario [...]: M/R/O [W][C]	Element	Value	[High-Level Mapping] Element and value rules
2: M 3: M 4: M 5: M 6: M 7: M	ElectricalConnection. electricalConnectionDescriptionListData. electricalConnectionDescriptionData.		
2: M 3: M 4: M 5: M 6: M 7: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M 3: M 4: M 5: M 6: M 7: M	powerSupplyType	"ac"	
2: M 3: M 4: M 5: M 6: M 7: M	positiveEnergyDirection	"consume"	

Table 28: Content of Function "electricalConnectionDescriptionListData" at Actor Grid Connection Point

3.2.2.2.4.2 Function "electricalConnectionParameterDescriptionListData"

Scenario [...]: M/R/O [W][C]	Element	Value	[High-Level Mapping] Element and Value rules
2: M	ElectricalConnection. electricalConnectionParameterDescriptionListData. electricalConnectionParameterDescriptionData.		

2: M	electricalConnectionId	<ec1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
2: M	parameterId	<p1#{1..1}->ec1#1>	SHALL be set as SUB IDENTIFIER.
2: M	measurementId	<m1#{1..1}->p1#1>	SHALL be set as FOREIGN IDENTIFIER.
2: M	voltageType	"ac"	
2: O	acMeasuredPhases	"abc" "ab" "bc" "ac" "a" "b" "c" (->p1#1)	If the Monitored Unit is connected to less than three phases, one of the other combinations like "a" or "ab" are allowed instead of "abc". The values "a", "b", and "c" are permitted if and only if only one phase is connected to the Monitored Unit.
2: O	acMeasuredInReferenceTo	"neutral"	
2: M	acMeasurementType	"real"	
2: O	acMeasurementVariant	"rms"	
3: M	ElectricalConnection. electricalConnectionParameterDescriptionListData. electricalConnectionParameterDescriptionData.		
3: M	electricalConnectionId	<ec1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
3: M	parameterId	<p2#{1..1}->ec1#1>	SHALL be set as SUB IDENTIFIER.
3: M	measurementId	<m2#{1..1}->p2#1>	SHALL be set as FOREIGN IDENTIFIER.
3: M	voltageType	"ac"	
3: M	acMeasurementType	"real"	
4: M	ElectricalConnection. electricalConnectionParameterDescriptionListData. electricalConnectionParameterDescriptionData.		
4: M	electricalConnectionId	<ec1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
4: M	parameterId	<p3#{1..1}->ec1#1>	SHALL be set as SUB IDENTIFIER.
4: M	measurementId	<m3#{1..1}->p3#1>	SHALL be set as FOREIGN IDENTIFIER.
4: M	voltageType	"ac"	
4: M	acMeasurementType	"real"	
5: M	ElectricalConnection. electricalConnectionParameterDescriptionListData. electricalConnectionParameterDescriptionData.		
5: M	electricalConnectionId	<ec1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
5: M	parameterId	<p4#{1..3}->ec1#1>	SHALL be set as SUB IDENTIFIER.
5: M	measurementId	<m4#{1..3}->p4#{1..3}>	SHALL be set as FOREIGN IDENTIFIER.
5: M	voltageType	"ac"	
5: M	acMeasuredPhases	"a" (-><p4#1>)*1	[MGCP-051/1]
		"b" (-><p4#2>)*1	[MGCP-051/2]
		"c" (-><p4#3>)*1	[MGCP-051/3]
5: M	acMeasurementType	"real"	[MGCP-050/1]
6: M	acMeasurementVariant	"rms"	[MGCP-050/2]
6: M	ElectricalConnection. electricalConnectionParameterDescriptionListData. electricalConnectionParameterDescriptionData.		

6: M	electricalConnectionId	<ec1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
6: M	parameterId	<p5#{1..6}->ec1#1>	SHALL be set as SUB IDENTIFIER.
6: M	measurementId	<m5#{1..6}->p5#{1..6}>	SHALL be set as FOREIGN IDENTIFIER.
6: M	voltageType	"ac"	
6: M	acMeasuredPhases	"a" (-><p5#1>)	[MGCP-061/1]
		"a" (-><p5#4>)	[MGCP-061/4]
		"b" (-><p5#2>)	[MGCP-061/2]
		"b" (-><p5#5>)	[MGCP-061/5]
		"c" (-><p5#3>)	[MGCP-061/3]
		"c" (-><p5#6>)	[MGCP-061/6]
6: M	acMeasuredInReferenceTo	"a" (-><p5#6>)	[MGCP-061/6]
		"b" (-><p5#4>)	[MGCP-061/4]
		"c" (-><p5#5>)	[MGCP-061/5]
		"neutral" (-><p5#1>)	[MGCP-061/1]
		"neutral" (-><p5#2>)	[MGCP-061/2]
		"neutral" (-><p5#3>)	[MGCP-061/3]
6: M	acMeasurementType	"apparent"	
6: M	acMeasurementVariant	"rms"	
7: M	ElectricalConnection. electricalConnectionParameterDescriptionListData. electricalConnectionParameterDescriptionData.		
7: M	electricalConnectionId	<ec1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
7: M	parameterId	<p6#{1..1}->ec1#1>	SHALL be set as SUB IDENTIFIER.
7: M	measurementId	<m6#{1..1}->p6#1>	SHALL be set as FOREIGN IDENTIFIER.
7: M	voltageType	"ac"	

Table 29: Content of Function "electricalConnectionParameterDescriptionListData" at Actor Grid Connection Point

*1: Each permitted value of the Element "acMeasuredPhases" SHALL NOT be used for more than one value of Element "parameterId".

3.2.2.3 Client data - Specializations

As this Actor has only server functionality, no Specializations are described within this section.

3.3 Pre-Scenario communication

3.3.1 General information

The Pre-Scenario communication is needed if a client does not know the corresponding addresses on the server or if the required subscriptions or bindings are not active. In this case certain general communication mechanisms SHALL be used within SPINE:

- Detailed discovery: allows to discover resource addresses.
- Binding: allows to bind to resource address, which is frequently necessary to obtain write privileges.

1038 c) Subscription: allows to subscribe to resource addresses, which is necessary to receive
1039 unsolicited notifications if a resource changes during runtime.

1040 It is possible to combine those steps for multiple Scenarios or also multiple Use Cases:

- 1041 - E.g. if multiple Scenarios in multiple Use Cases use the same Feature, only one subscription
1042 needs to occur.
- 1043 - E.g. a complete detailed discovery or a subscription to the NodeManagement Feature needs
1044 to occur only once for all Use Cases.

1045 Depending on which Entity, Feature and Functions are used within a Scenario the payload of the
1046 corresponding messages may slightly differ, but the basic principles and messages used stay the
1047 same.

1048 The subsequent messages SHALL be exchanged for those parts that have not already been performed
1049 since the current connection is established or if those parts are outdated for another reason (e.g. if
1050 the detailed discovery is needed, but the bindings and subscriptions are still active from a previous
1051 connection only the detailed discovery messages need to be exchanged). If all Pre-Scenario
1052 communication parts are up to date, this section MAY be skipped, and the implementation can
1053 proceed as described in the corresponding "Scenario communication" sections.

1054 After the connection is re-established (e.g. due to a power loss or a firmware update) the Pre-
1055 Scenario communication SHALL be performed as well. There may be circumstances where messages
1056 from the Pre-Scenario communication may be exchanged again.

1057 Often the necessary messages of different Scenarios can be combined, so that only one single
1058 message is needed instead of multiple messages for the different Scenarios. This also is the case for
1059 the Pre-Scenario communication. In most cases only one "read" operation on the detailed discovery
1060 is necessary, as well as only one subscription request or binding request is needed for each Feature.
1061 Often multiple Scenarios within a Use Case access the same Feature, so only one subscription or
1062 binding is necessary.

1063

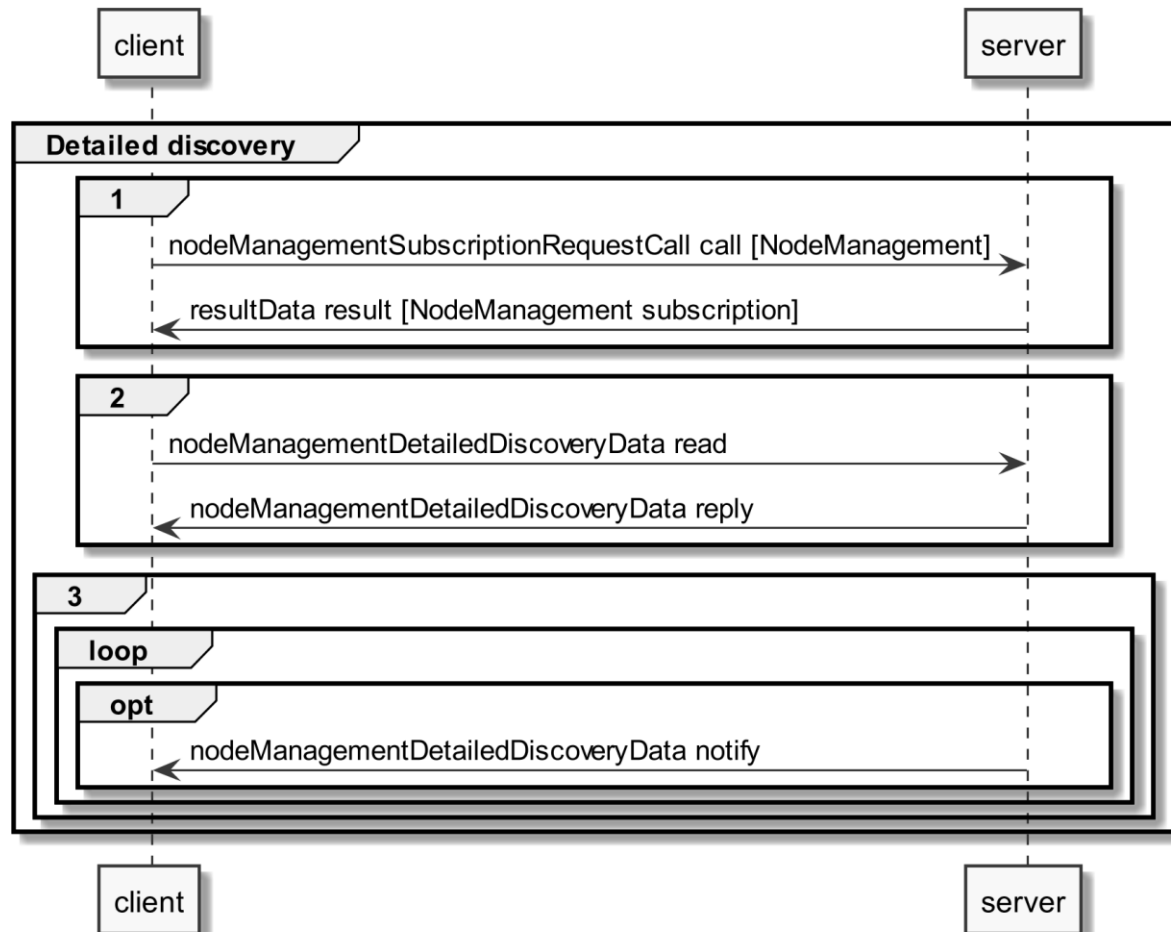
1064 3.3.2 Detailed discovery

1065 For the functionality where a client already has current detailed discovery information (i.e.
1066 independent of this Use Case or any Scenario of it) the remainder of this section SHOULD be skipped.

1067 Otherwise, the following procedure SHALL be performed in the given order:

- 1068 1. If a client is not subscribed to the primary NodeManagement instance, the client SHALL
1069 acquire a subscription according to the figure provided within this sub-section.
- 1070 2. A client SHALL read the detailed discovery information according to the figure provided
1071 within this sub-section. It SHALL keep the received information as far as it concerns
1072 mandatory and supported optional Entity Types, Feature Types and Functions of this Use
1073 Case that are needed by the client. This means that a client may choose how to store the
1074 necessary information. E.g. a client Actor can store the information how to address the
1075 necessary Features of the implemented Scenarios but may discard the Entity information.

- 1076 3. If and as long as a client has a subscription to the detailed discovery information of an Actor
 1077 and receives proper notifications, it SHALL consider (i.e. integrate into the kept detailed
 1078 discovery information) the received information as far as it concerns mandatory and
 1079 supported optional Entity Types, Feature Types and Functions of this Use Case.



1080

1081 *Figure 7: Pre-Scenario communication - Detailed discovery sequence diagram*

1082 If the "nodeManagementDetailedDiscoveryData read" fails, the client SHOULD retry to read the
 1083 detailed discovery information until the "nodeManagementDetailedDiscoveryData reply" message
 1084 was received successfully.

1085 If all functionality is present at all times: The "nodeManagementDetailedDiscoveryData reply"
 1086 message contains at least the mandatory Entities and Features given in the "Actor [...] overview"
 1087 diagrams as well as the used Functions and their "possible operations" described in section 3.2 and
 1088 its sub-sections.

1089 If functionality is added or removed dynamically: The "nodeManagementDetailedDiscoveryData
 1090 reply" message does not need to contain all mandatory Entities and Features given in the "Actor [...]
 1091 overview" diagrams as well as all needed Functions and their "possible operations" described in
 1092 section 3.2 and its sub-sections. However, as soon as the functionality is available it will be
 1093 announced via a "nodeManagementDetailedDiscoveryData notify" message.

1094 For the nodeManagementDetailedDiscoveryData read Function it is recommended to use a partial
 1095 read with separated Selectors that may use one of the following Elements:

- 1096 - entityType
- 1097 - featureType

1098 Note: Even with the usage of Selectors Features and Entities that are not relevant for this Use Case
 1099 may be discovered. However, only Features and Entities that fulfil the hierarchical order as described
 1100 within the Actors' sections shall be considered for this Use Case.

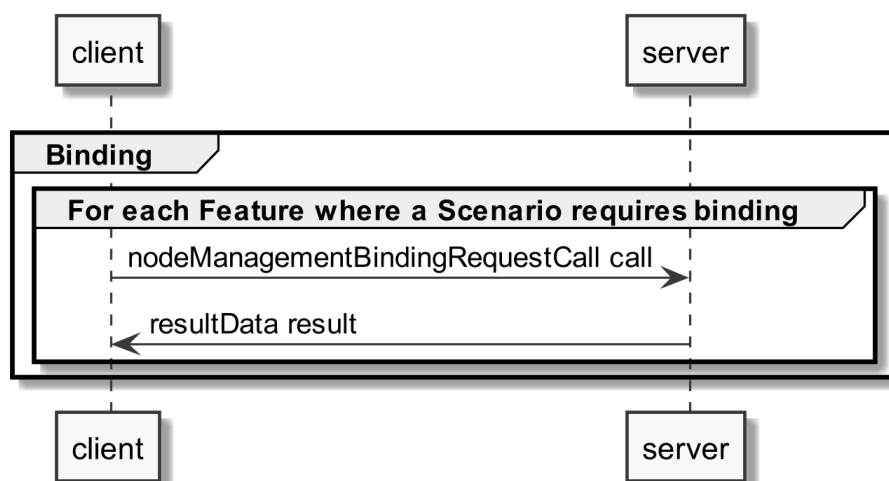
1101 A "partial" notify SHALL be supported without using Selectors and Elements. Partial "delete" notify
 1102 SHOULD also be supported with separated Selectors that may use one of the following Elements:

- 1103 - entityAddress
- 1104 - featureAddress

1105

1106 3.3.3 Binding

1107 If binding is required by a Scenario that uses Features with writeable or changeable data, the server
 1108 SHALL support binding for the respective Features. Before a write on a Function of a Feature occurs,
 1109 the client SHALL create a binding to the corresponding Feature. For this the
 1110 nodeManagementBindingRequestCall Function is used as shown in the following sequence diagram:



1111

1112 *Figure 8: Pre-Scenario communication - Binding sequence diagram*

1113 If functionality is added or removed dynamically, binding may not be possible at all times on the
 1114 required Functions. A client SHALL retry to create a binding again when receiving according updated
 1115 detailed discovery information.

1116

1117 3.3.4 Subscription

1118 A server SHALL support subscription for all Features that contain readable data that may change
 1119 during runtime. The client SHALL create a subscription for all Features that the client wants to read.
 1120 For this the nodeManagementSubscriptionRequestCall Function is used as shown in the following
 1121 sequence diagram:

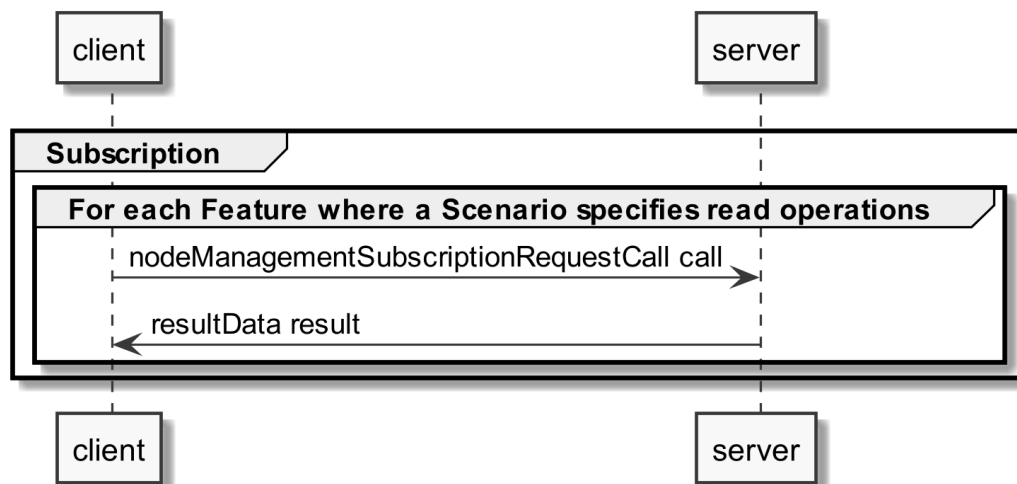


Figure 9: Pre-Scenario communication - Subscription sequence diagram

If the subscription request fails (e.g. because it is not supported by the server or the maximum number of possible subscriptions is reached), the client SHOULD read the data periodically (so-called "polling").

If functionality is added or removed dynamically, subscription may not be possible at all times on the required Functions. A client SHALL retry its subscription procedure again when receiving according updated detailed discovery information.

3.3.5 Dynamic behaviour

In case Entities or Features are removed, a nodeManagementDetailedDiscoveryData "notify" is transmitted that informs about the deleted Entities and Features. All existing binding or subscription entries on the deleted Features SHALL be deleted by each device.

In case Entities or Features are added the Pre-Scenario communication starts with transmitting a nodeManagementDetailedDiscoveryData "notify" that contains the added Entities and Features.

3.4 Scenarios

3.4.1 Scenario 1 - Monitor PV feed-in power limitation factor

3.4.1.1 Pre-Scenario communication

1. **Detailed discovery:** Actors that act as client within this Scenario, need to know the addresses of the server Features used in the Initial Scenario communication. If the address of a particular server Feature is not known, the detailed discovery must be used, as described in section 3.3.2.
2. **Binding:** Binding SHOULD NOT be used for this Scenario.
3. **Subscription:** Actors SHALL create a subscription for each server Feature that is relevant for the corresponding Actor within this Scenario, as described in section 3.3.4.

The Initial Scenario communication SHALL start at the latest when the required resources on an Actor are known and the necessary binding and subscription procedures have been finished. However, as

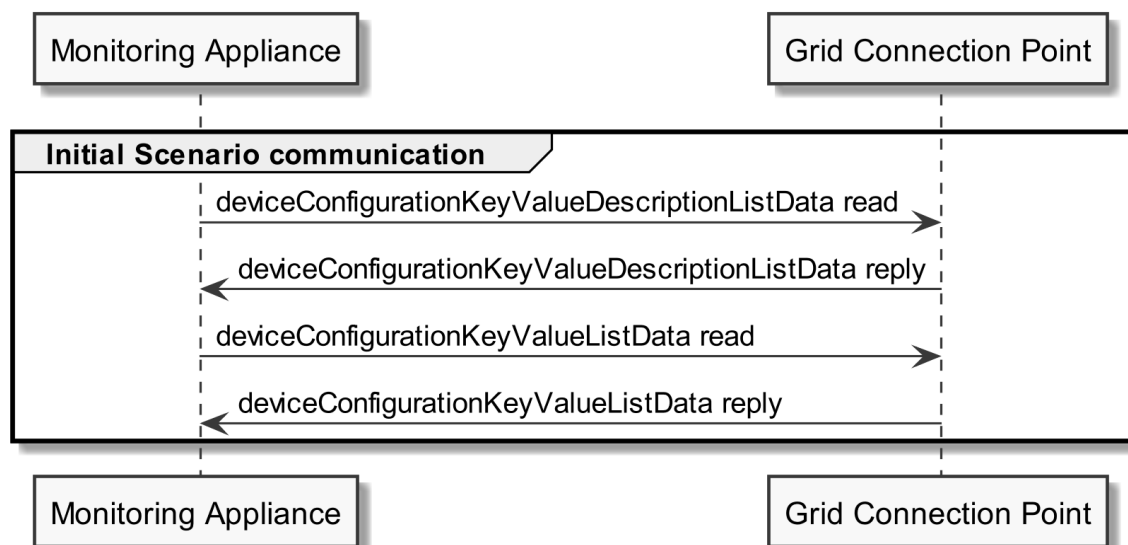
1150 soon as the address of a required resource is known, the Initial Scenario communication for this
 1151 resource MAY start already, even if the addresses of other required resources are not known yet.

1152 If required resources are removed and added again, they are re-discovered, and the Initial Scenario
 1153 communication is triggered again for those resources.

1154

1155 **3.4.1.2 Initial Scenario communication**

1156 Each time a (re-)connection is established, even if the Pre-Scenario communication phase is skipped,
 1157 the messages shown in the following sequence diagram SHALL be exchanged, as the corresponding
 1158 resources may have changed in the meantime:



1159

1160 *Figure 10: Scenario 1 - Initial Scenario communication sequence diagram*

1161 The deviceConfigurationKeyValueDescriptionListData read SHOULD be a "partial" read operation
 1162 with the following Selectors:

- 1163 - keyName = "pvCurtailementLimitFactor"

1164 The deviceConfigurationKeyValueListData read SHOULD be a "partial" read operation with the
 1165 following Selectors:

- 1166 - keyId (derived from the deviceConfigurationKeyValueDescriptionListData reply)

1167 Note: If partial read is not supported a full read SHALL be performed.

1168

1169 The following table shows where the required content of the messages from the sequence diagram is
 1170 described:

Message name from sequence diagram	Content description in table	Scenario number in table
deviceConfigurationKeyValueDescriptionListData reply	Table 23	1
deviceConfigurationKeyValueListData reply	Table 24	1

1171 *Table 30: Initial Scenario communication content references for Scenario 1*

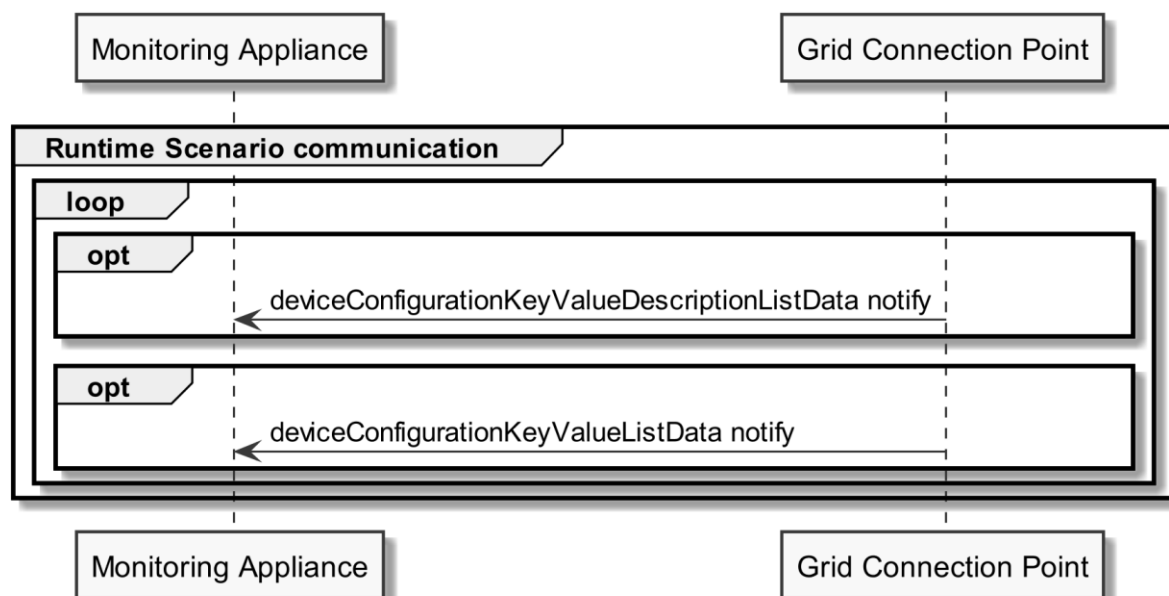
1172 Note: Within the Initial Scenario communication, the content required by this Scenario MAY not be
 1173 provided completely, but later during Runtime Scenario communication.

1174

1175 **3.4.1.3 Runtime Scenario communication**

1176 Based on the Initial Scenario communication, the Runtime Scenario communication provides updates
 1177 during runtime.

1178 If one of the referenced server Functions' data change, the server SHALL submit the change as shown
 1179 in the following figure:



1180

1181 *Figure 11: Scenario 1 - Runtime Scenario communication sequence diagram*

1182 Note: Normally, in this Scenario the configuration parameter does not change during runtime. Hence,
 1183 usually no notifications are sent during runtime in this Scenario.

1184 Partial notifications without Selectors or Elements SHALL be supported for all Functions used in this
 1185 Scenario.

1186 For deviceConfigurationKeyValueDescriptionListData notify and
 1187 deviceConfigurationKeyValueListData notify "partial" delete notifications SHOULD be supported with
 1188 the Selector:

1189 - keyId

1190 Note: To interpret partial notification messages correctly, the information obtained during the Initial
 1191 Scenario communication phase is required.

1192 Note: A read operation ("polling") on all Functions is possible at any time, e.g. if a notification could
 1193 not be evaluated.

1194

1195 The following table shows where the required content of the messages of the sequence diagram is
 1196 described:

Message name from sequence diagram	Content description in table	Scenario number in table
deviceConfigurationKeyValueDescriptionListData notify	Table 23	1
deviceConfigurationKeyValueListData notify	Table 24	1

1197 *Table 31: Runtime Scenario communication content references for Scenario 1*

1198

1199 **3.4.1.4 Additional information**

1200 None.

1201

1202 **3.4.2 Scenario 2 - Monitor momentary power consumption/production**

1203 **3.4.2.1 Pre-Scenario communication**

- 1204 1. **Detailed discovery:** Actors that act as client within this Scenario, need to know the addresses
 1205 of the server Features used in the Initial Scenario communication. If the address of a
 1206 particular server Feature is not known, the detailed discovery must be used, as described in
 1207 section 3.3.2.
- 1208 2. **Binding:** Binding SHOULD NOT be used for this Scenario.
- 1209 3. **Subscription:** Actors SHALL create a subscription for each server Feature that is relevant for
 1210 the corresponding Actor within this Scenario, as described in section 3.3.4.

1211 The Initial Scenario communication SHALL start at the latest when the required resources on an Actor
 1212 are known and the necessary binding and subscription procedures have been finished. However, as
 1213 soon as the address of a required resource is known, the Initial Scenario communication for this
 1214 resource MAY start already, even if the addresses of other required resources are not known yet.

1215 If required resources are removed and added again, they are re-discovered, and the Initial Scenario
 1216 communication is triggered again for those resources.

1217

1218 **3.4.2.2 Initial Scenario communication**

1219 Each time a (re-)connection is established, even if the Pre-Scenario communication phase is skipped,
 1220 the messages shown in the following sequence diagram SHALL be exchanged, as the corresponding
 1221 resources may have changed in the meantime:

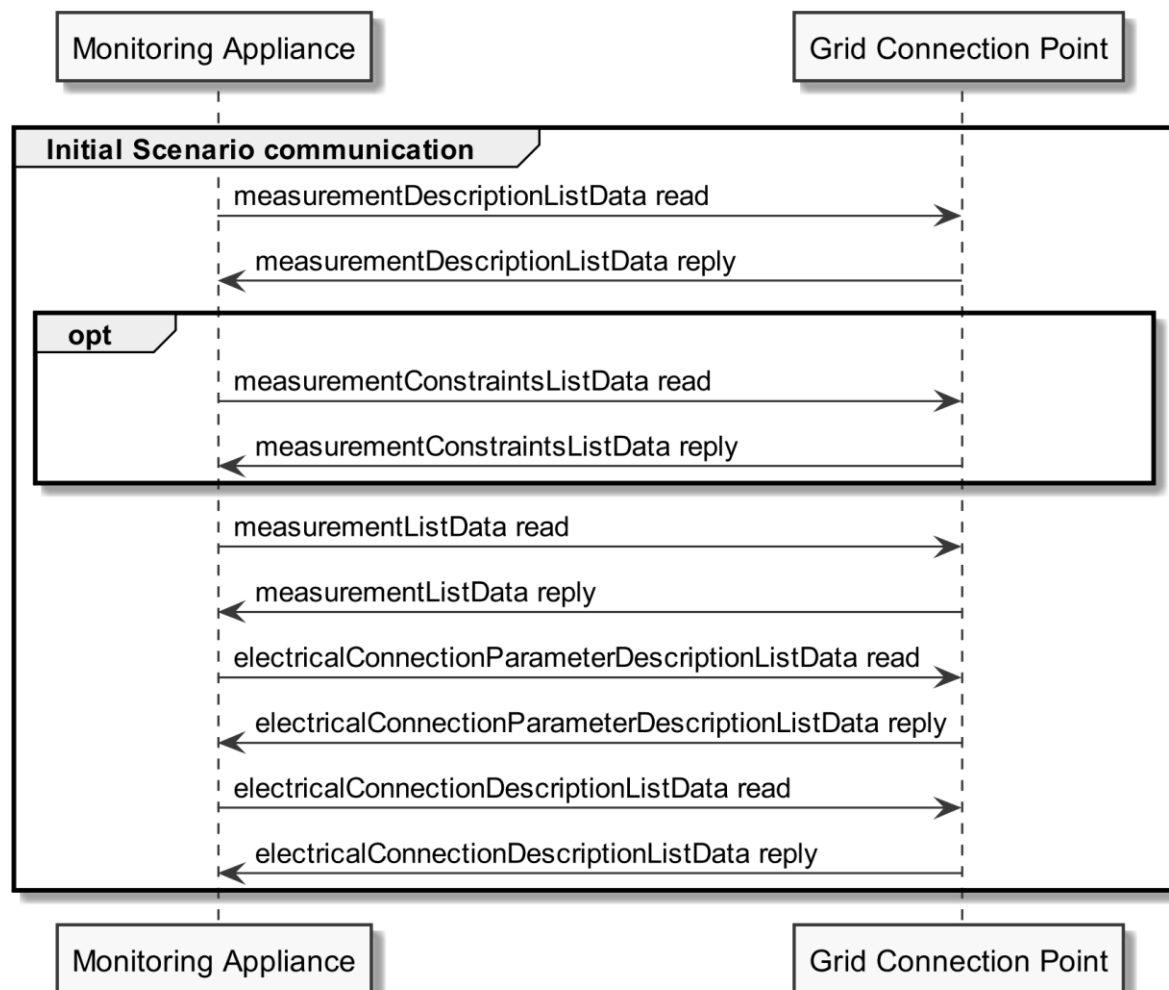


Figure 12: Scenario 2 - Initial Scenario communication sequence diagram

Note: The initiation of the optional sequence part (marked with "opt" in the figure) is optional for the "Monitoring Appliance" even if the "Grid Connection Point" can provide the requested Function.

The `measurementDescriptionListData read` SHOULD be a "partial" read operation with the following Selectors:

- `scopeType = "acPowerTotal"`

The `measurementConstraintsListData read`, `measurementListData read` and `electricalConnectionParameterDescriptionListData read` SHOULD be a "partial" read operation with the following Selectors:

- `measurementId` (derived from the `measurementDescriptionListData reply`)

The `electricalConnectionDescriptionListData read` SHOULD be a "partial" read operation with the following Selectors:

- `electricalConnectionId` (derived from the `electricalConnectionParameterDescriptionListData reply`)

Note: If partial read is not supported a full read SHALL be performed.

1238

1239 The following table shows where the required content of the messages of the sequence diagram is
1240 described:

Message name from sequence diagram	Content description in table	Scenario number in table
measurementDescriptionListData reply	Table 25	2
measurementConstraintsListData reply	Table 26	2
measurementListData reply	Table 27	2
electricalConnectionParameterDescriptionListData reply	Table 29	2
electricalConnectionDescriptionListData reply	Table 28	2

1241 *Table 32: Initial Scenario communication content references for Scenario 2*

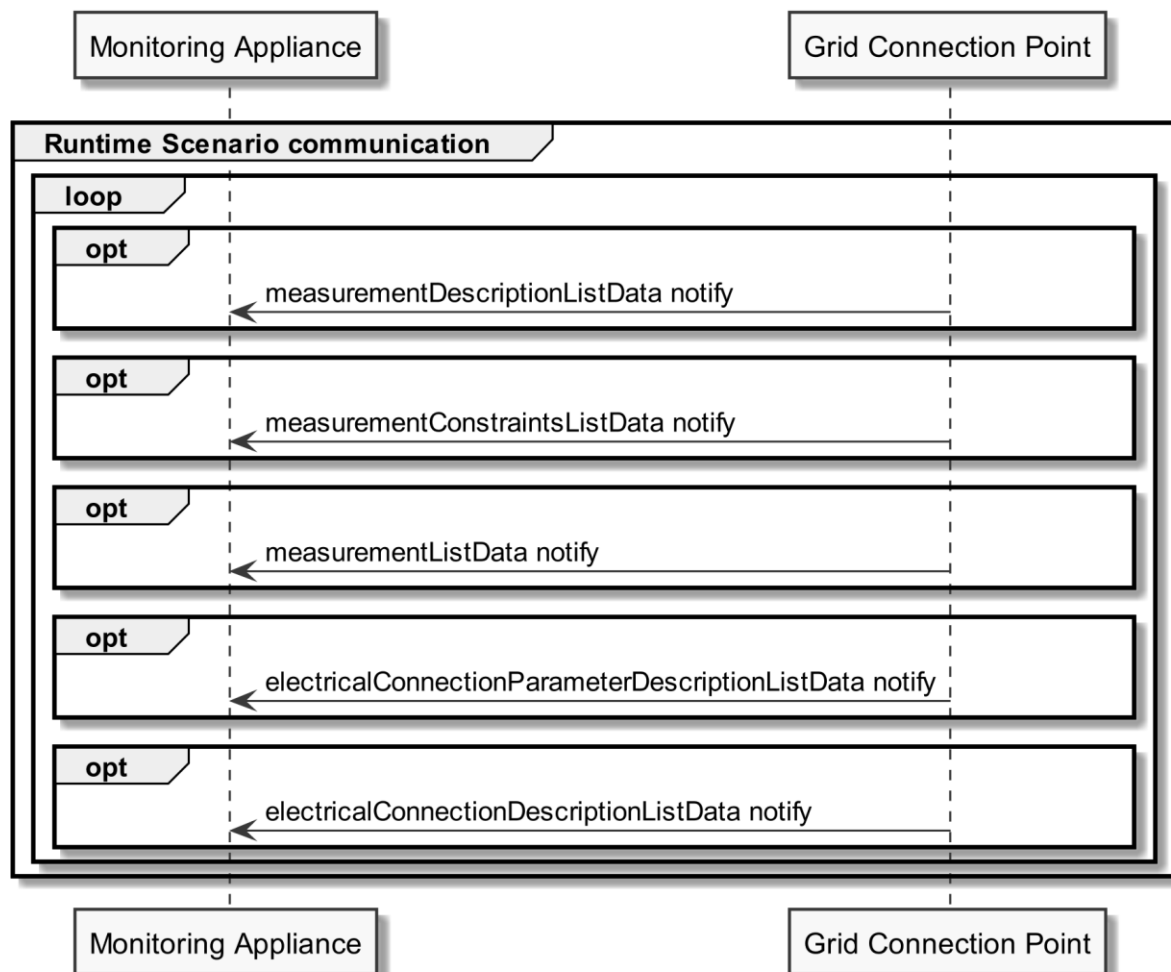
1242 Note: Within the Initial Scenario communication, the content required by this Scenario MAY not be
1243 provided completely, but later during Runtime Scenario communication.

1244

1245 **3.4.2.3 Runtime Scenario communication**

1246 Based on the Initial Scenario communication, the Runtime Scenario communication provides updates
1247 during runtime.

1248 If one of the referenced server Functions' data change the server SHALL submit the change as shown
1249 in the following figure:



1250

1251 *Figure 13: Scenario 2 - Runtime Scenario communication sequence diagram*

1252 Note: Normally, in this Scenario only the "measurementListData" Function changes during runtime.
 1253 Hence, usually no notifications of the other Functions of this Scenario are sent during runtime.

1254 Partial notifications without Selectors or Elements SHALL be supported for all Functions used in this
 1255 Scenario.

1256 For measurementDescriptionListData notify, measurementConstraintsListData notify and
 1257 measurementListData notify "partial" delete notifications SHOULD be supported with the Selector:

1258 - measurementId

1259 For electricalConnectionParameterDescriptionListData notify "partial" delete notifications SHOULD
 1260 be supported with the Selectors:

1261 - electricalConnectionId

1262 - parameterId

1263 - measurementId

1264 Note: To interpret partial notification messages correctly, the information obtained during the Initial
 1265 Scenario communication phase is required.

Note: A read operation ("polling") on all Functions is possible at any time, e.g. if a notification could not be evaluated.

The following table shows where the required content of the messages of the sequence diagram is described:

Message name from sequence diagram	Content description in table	Scenario number in table
measurementDescriptionListData notify	Table 25	2
measurementConstraintsListData notify	Table 26	2
measurementListData notify	Table 27	2
electricalConnectionParameterDescriptionListData notify	Table 29	2
electricalConnectionDescriptionListData notify	Table 28	2

Table 33: Runtime Scenario communication content references for Scenario 2

3.4.2.4 Additional information

Note: Both, consumed and produced power, are represented by the single measurement value of this Scenario. Whether the device consumes or produces power is indicated by the sign of the measurement value: The Element "positiveEnergyDirection" in the Function "electricalConnectionDescriptionListData" determines how the sign SHALL be interpreted. Within this Use Case the Element has a fixed value of "consume". Hence, the "passive sign convention" is applied and therefore positive values SHALL be interpreted as consumed power and negative values SHALL be interpreted as produced power (see section 2.6.1 for details).

Note: Within this Use Case, only the newest measurement value SHALL be stated ([MGCP-004]). Additional historical values are forbidden.

3.4.3 Scenario 3 - Monitor total feed-in energy

3.4.3.1 Pre-Scenario communication

- Detailed discovery:** Actors that act as client within this Scenario, need to know the addresses of the server Features used in the Initial Scenario communication. If the address of a particular server Feature is not known, the detailed discovery must be used, as described in section 3.3.2.
- Binding:** Binding SHOULD NOT be used for this Scenario.
- Subscription:** Actors SHALL create a subscription for each server Feature that is relevant for the corresponding Actor within this Scenario, as described in section 3.3.4.

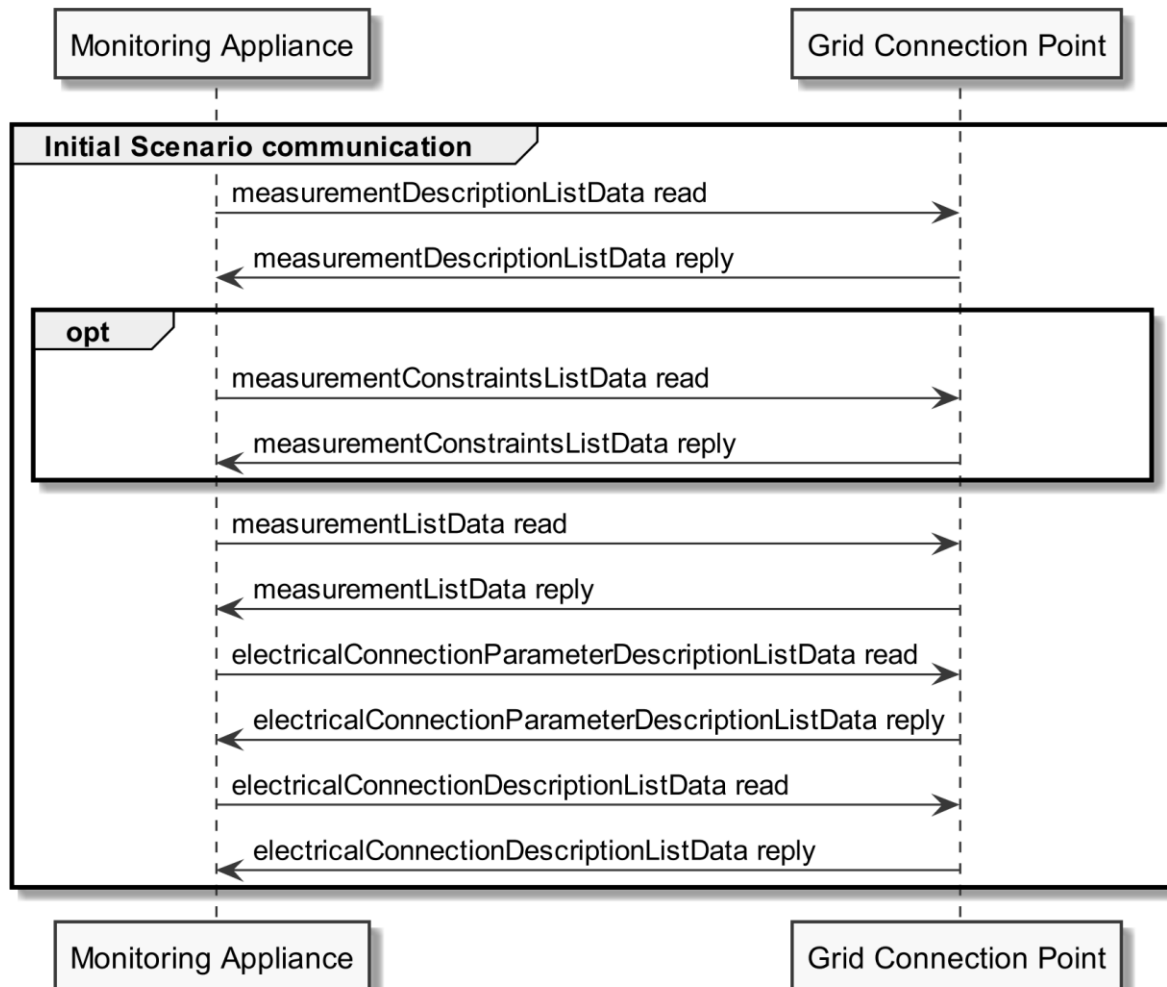
The Initial Scenario communication SHALL start at the latest when the required resources on an Actor are known and the necessary binding and subscription procedures have been finished. However, as soon as the address of a required resource is known, the Initial Scenario communication for this resource MAY start already, even if the addresses of other required resources are not known yet.

1297 If required resources are removed and added again, they are re-discovered, and the Initial Scenario
 1298 communication is triggered again for those resources.

1299

1300 3.4.3.2 Initial Scenario communication

1301 Each time a (re-)connection is established, even if the Pre-Scenario communication phase is skipped,
 1302 the messages shown in the following sequence diagram SHALL be exchanged, as the corresponding
 1303 resources may have changed in the meantime:



1304

1305 *Figure 14: Scenario 3 - Initial Scenario communication sequence diagram*

1306 Note: The initiation of the optional sequence part (marked with "opt" in the figure) is optional for the
 1307 "Monitoring Appliance" even if the "Grid Connection Point" can provide the requested Function.

1308 The measurementDescriptionListData read SHOULD be a "partial" read operation with the following
 1309 Selectors:

- 1310 - scopeType = "gridFeedIn"

1311 The measurementConstraintsListData read, measurementListData read and
 1312 electricalConnectionParameterDescriptionListData read SHOULD be a "partial" read operation with
 1313 the following Selectors:

1314 - measurementId (derived from the measurementDescriptionListData reply)

1315 The electricalConnectionDescriptionListData read SHOULD be a "partial" read operation with the
1316 following Selectors:

1317 - electricalConnectionId (derived from the electricalConnectionParameterDescriptionListData
1318 reply)

1319 Note: If partial read is not supported a full read SHALL be performed.

1320

1321 The following table shows where the required content of the messages of the sequence diagram is
1322 described:

Message name from sequence diagram	Content description in table	Scenario number in table
measurementDescriptionListData reply	Table 25	3
measurementConstraintsListData reply	Table 26	3
measurementListData reply	Table 27	3
electricalConnectionParameterDescriptionListData reply	Table 29	3
electricalConnectionDescriptionListData reply	Table 28	3

1323 *Table 34: Initial Scenario communication content references for Scenario 3*

1324 Note: Within the Initial Scenario communication, the content required by this Scenario MAY not be
1325 provided completely, but later during Runtime Scenario communication.

1326

1327 **3.4.3.3 Runtime Scenario communication**

1328 Based on the Initial Scenario communication, the Runtime Scenario communication provides updates
1329 during runtime.

1330 If one of the referenced server Functions' data change the server SHALL submit the change as shown
1331 in the following figure:

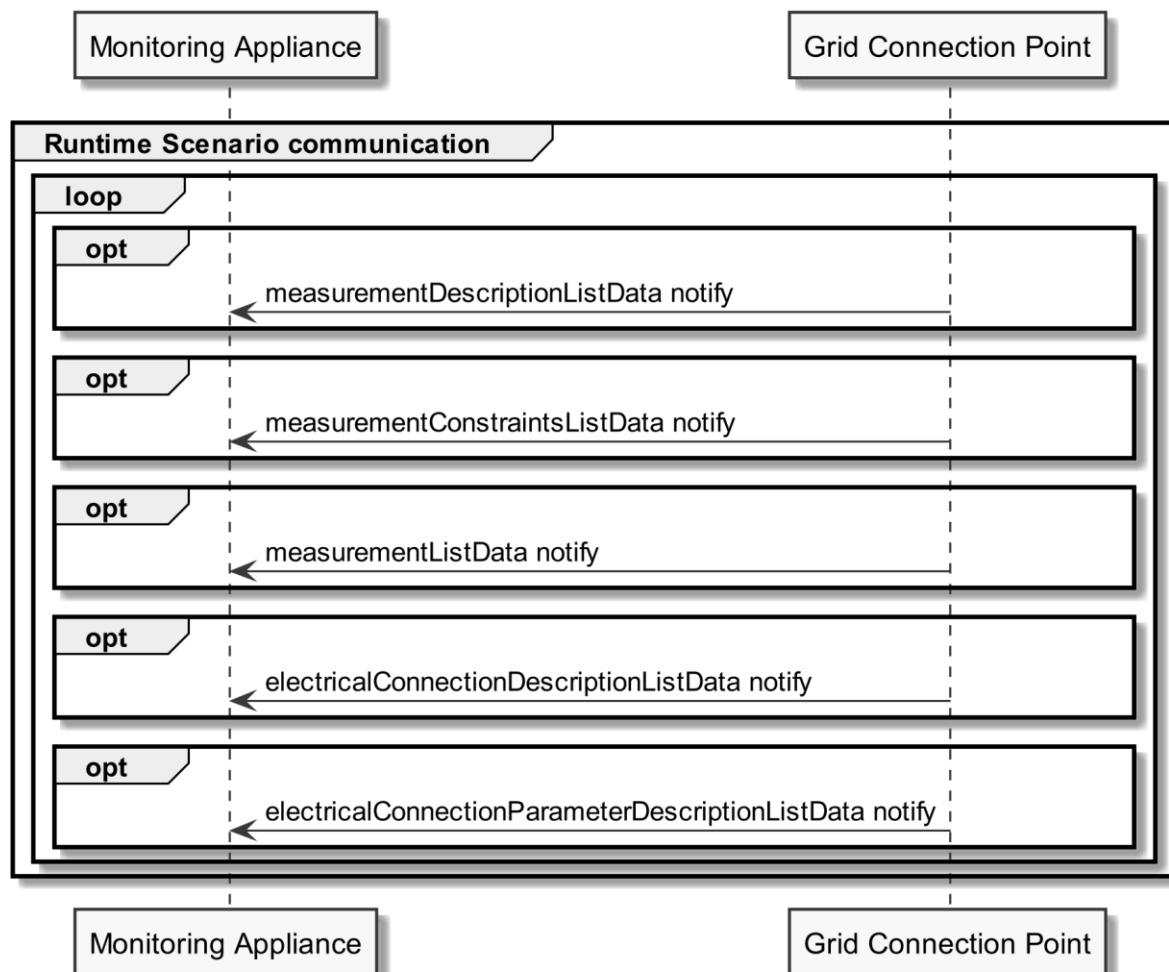


Figure 15: Scenario 3 - Runtime Scenario communication sequence diagram

Note: Normally, in this Scenario only the "measurementListData" Function changes during runtime. Hence, usually no notifications of the other Functions of this Scenario are sent during runtime.

Partial notifications without Selectors or Elements SHALL be supported for all Functions used in this Scenario.

For measurementDescriptionListData notify, measurementConstraintsListData notify and measurementListData notify "partial" delete notifications SHOULD be supported with the Selector:

- measurementId

For electricalConnectionParameterDescriptionListData notify "partial" delete notifications SHOULD be supported with the Selectors:

- electricalConnectionId
- parameterId
- measurementId

Note: To interpret partial notification messages correctly, the information obtained during the Initial Scenario communication phase is required.

Note: A read operation ("polling") on all Functions is possible at any time, e.g. if a notification could not be evaluated.

The following table shows where the required content of the messages from the sequence diagram is described:

Message name from sequence diagram	Content description in table	Scenario number in table
measurementDescriptionListData notify	Table 25	3
measurementConstraintsListData notify	Table 26	3
measurementListData notify	Table 27	3
electricalConnectionParameterDescriptionListData notify	Table 29	3
electricalConnectionDescriptionListData notify	Table 28	3

Table 35: Runtime Scenario communication content references for Scenario 3

3.4.3.4 Additional information

Note: The feed-in energy is cumulated since the installation of the electrical smart meter or submeter or since manual change by the operator (e.g. manual reset of the value or firmware update, etc.). The total feed-in energy only counts energy fed into the grid and is not reduced by energy consumed from the grid [MGCP-007].

Note: Within this Use Case, only the newest measurement value SHALL be stated ([MGCP-006]). Additional historical values are forbidden.

3.4.4 Scenario 4 - Monitor total consumed energy

3.4.4.1 Pre-Scenario communication

- Detailed discovery:** Actors that act as client within this Scenario, need to know the addresses of the server Features used in the Initial Scenario communication. If the address of a particular server Feature is not known, the detailed discovery must be used, as described in section 3.3.2.
- Binding:** Binding SHOULD NOT be used for this Scenario.
- Subscription:** Actors SHALL create a subscription for each server Feature that is relevant for the corresponding Actor within this Scenario, as described in section 3.3.4.

The Initial Scenario communication SHALL start at the latest when the required resources on an Actor are known and the necessary binding and subscription procedures have been finished. However, as soon as the address of a required resource is known, the Initial Scenario communication for this resource MAY start already, even if the addresses of other required resources are not known yet.

If required resources are removed and added again, they are re-discovered, and the Initial Scenario communication is triggered again for those resources.

3.4.4.2 Initial Scenario communication

Each time a (re-)connection is established, even if the Pre-Scenario communication phase is skipped, the messages shown in the following sequence diagram SHALL be exchanged, as the corresponding resources may have changed in the meantime:

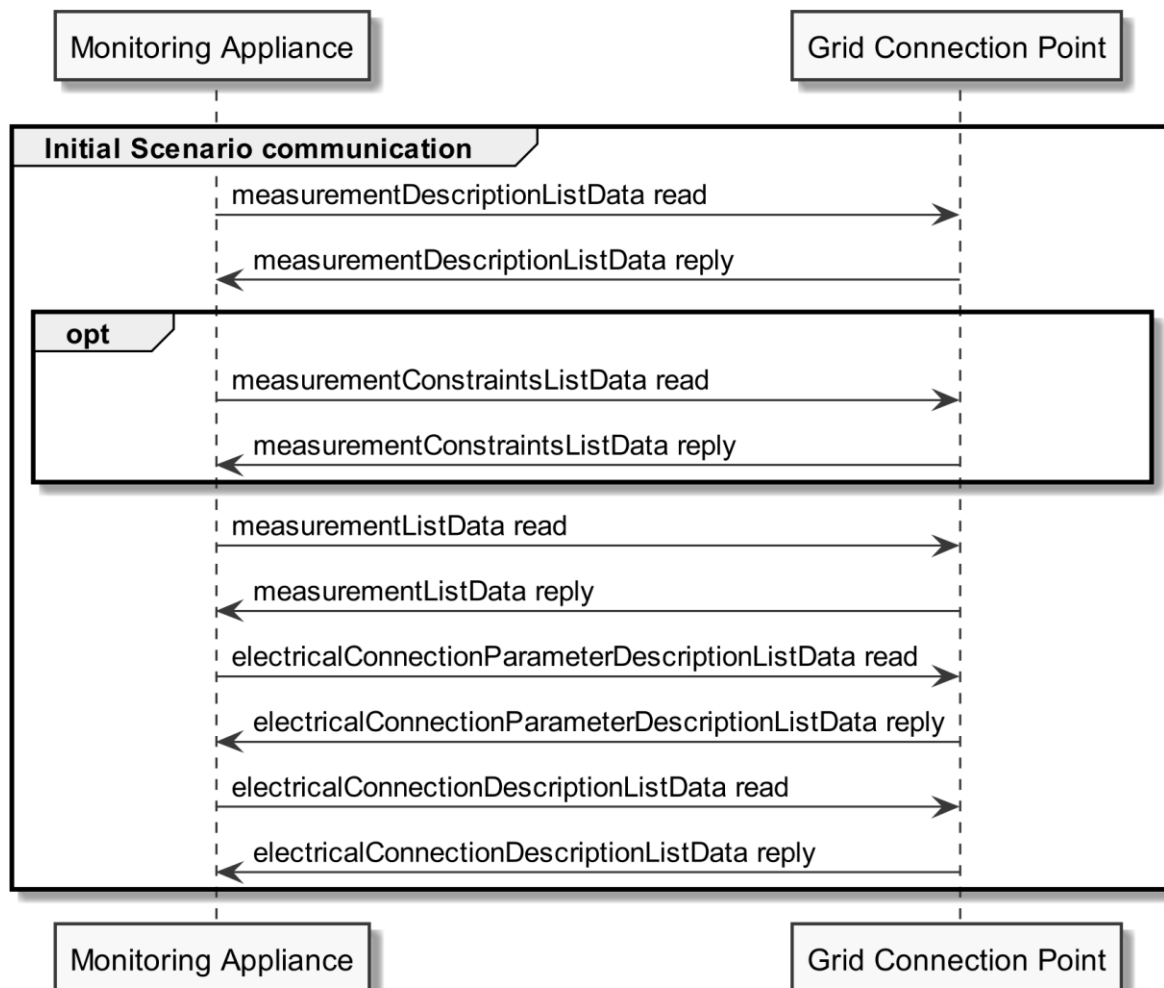


Figure 16: Scenario 4 - Initial Scenario communication sequence diagram

Note: The initiation of the optional sequence part (marked with "opt" in the figure) is optional for the "Monitoring Appliance" even if the "Grid Connection Point" can provide the requested Function.

The `measurementDescriptionListData read` SHOULD be a "partial" read operation with the following Selectors:

- `scopeType = "gridConsumption"`

The `measurementConstraintsListData read`, `measurementListData read` and `electricalConnectionParameterDescriptionListData read` SHOULD be a "partial" read operation with the following Selectors:

- `measurementId` (derived from the `measurementDescriptionListData reply`)

The `electricalConnectionDescriptionListData read` SHOULD be a "partial" read operation with the following Selectors:

1396 - electricalConnectionId (derived from the electricalConnectionParameterDescriptionListData
1397 reply)

1398 Note: If partial read is not supported a full read SHALL be performed.

1399

1400 The following table shows where the required content of the messages from the sequence diagram is
1401 described:

Message name from sequence diagram	Content description in table	Scenario number in table
measurementDescriptionListData reply	Table 25	4
measurementConstraintsListData reply	Table 26	4
measurementListData reply	Table 27	4
electricalConnectionDescriptionListData reply	Table 29	4
electricalConnectionDescriptionListData reply	Table 28	4

1402 *Table 36: Initial Scenario communication content references for Scenario 4*

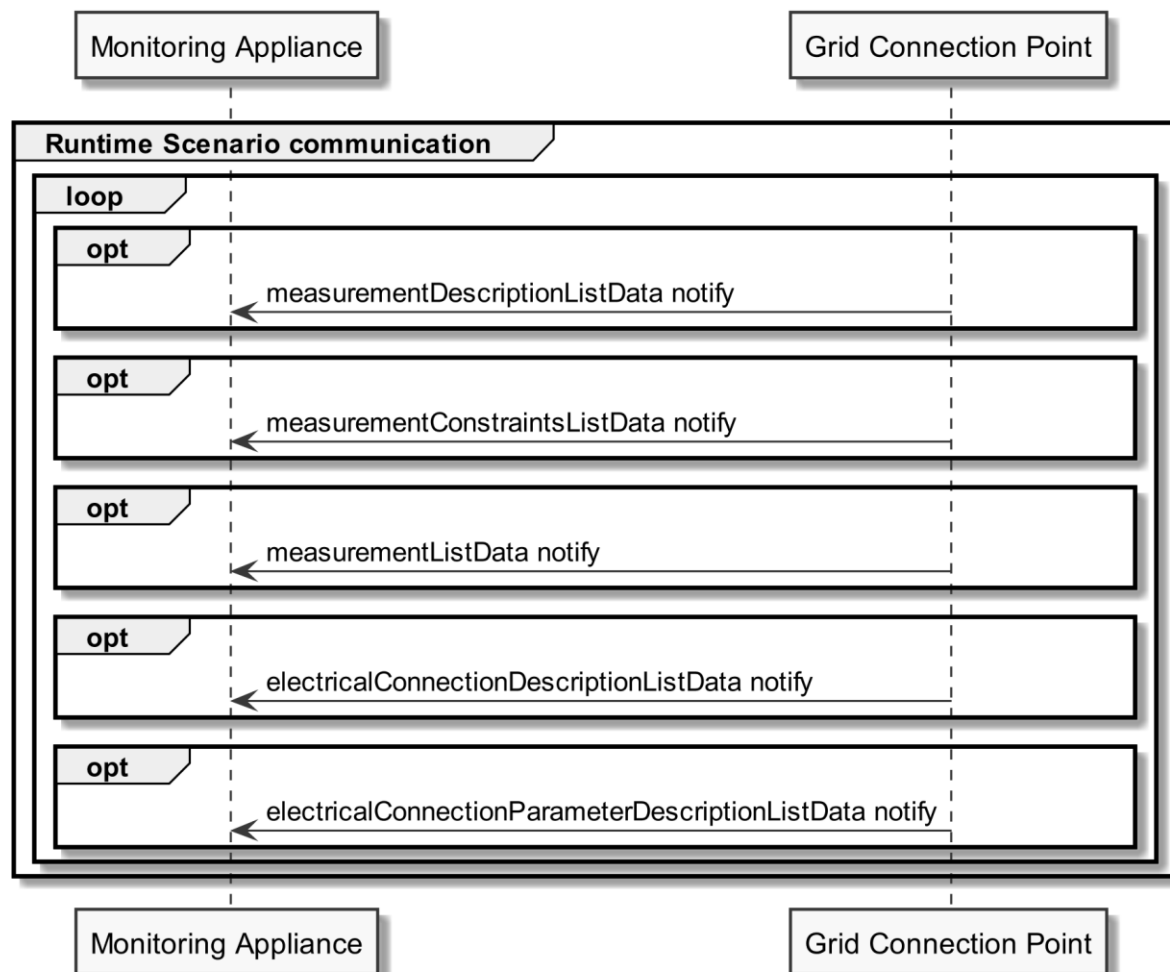
1403 Note: Within the Initial Scenario communication, the content required by this Scenario MAY not be
1404 provided completely, but later during Runtime Scenario communication.

1405

1406 **3.4.4.3 Runtime Scenario communication**

1407 Based on the Initial Scenario communication, the Runtime Scenario communication provides updates
1408 during runtime.

1409 If one of the referenced server Functions' data change the server SHALL submit the change as shown
1410 in the following figure:



1411

1412 *Figure 17: Scenario 4 - Runtime Scenario communication sequence diagram*

1413 Note: Normally, in this Scenario only the "measurementListData" Function changes during runtime.
 1414 Hence, usually no notifications of the other Functions of this Scenario are sent during runtime.

1415 Partial notifications without Selectors or Elements SHALL be supported for all Functions used in this
 1416 Scenario.

1417 For measurementDescriptionListData notify, measurementConstraintsListData notify and
 1418 measurementListData notify "partial" delete notifications SHOULD be supported with the Selector:

1419 - measurementId

1420 For electricalConnectionParameterDescriptionListData notify "partial" delete notifications SHOULD
 1421 be supported with the Selectors:

1422 - electricalConnectionId

1423 - parameterId

1424 - measurementId

1425 Note: To interpret partial notification messages correctly, the information obtained during the Initial
 1426 Scenario communication phase is required.

Note: A read operation ("polling") on all Functions is possible at any time, e.g. if a notification could not be evaluated.

The following table shows where the required content of the messages from the sequence diagram is described:

Message name from sequence diagram	Content description in table	Scenario number in table
measurementDescriptionListData notify	Table 25	4
measurementConstraintsListData notify	Table 26	4
measurementListData notify	Table 27	4
electricalConnectionParameterDescriptionListData notify	Table 29	4
electricalConnectionDescriptionListData notify	Table 28	4

Table 37: Runtime Scenario communication content references for Scenario 4

3.4.4.4 Additional information

Note: The consumed energy is cumulated since the installation of the electrical smart meter or submeter or since manual change by the operator (e.g. manual reset of the value or firmware update, etc.). The total consumed energy only counts energy consumed from the grid and is not reduced by energy fed into the grid [MGCP-008].

Note: Within this Use Case, only the newest measurement value SHALL be stated ([MGCP-006]). Additional historical values are forbidden.

3.4.5 Scenario 5 - Monitor momentary current consumption/production phase details

3.4.5.1 Pre-Scenario communication

- Detailed discovery:** Actors that act as client within this Scenario, need to know the addresses of the server Features used in the Initial Scenario communication. If the address of a particular server Feature is not known, the detailed discovery must be used, as described in section 3.3.2.
- Binding:** Binding SHOULD NOT be used for this Scenario.
- Subscription:** Actors SHALL create a subscription for each server Feature that is relevant for the corresponding Actor within this Scenario, as described in section 3.3.4.

The Initial Scenario communication SHALL start at the latest when the required resources on an Actor are known and the necessary binding and subscription procedures have been finished. However, as soon as the address of a required resource is known, the Initial Scenario communication for this resource MAY start already, even if the addresses of other required resources are not known yet.

If required resources are removed and added again, they are re-discovered, and the Initial Scenario communication is triggered again for those resources.

3.4.5.2 Initial Scenario communication

Each time a (re-)connection is established, even if the Pre-Scenario communication phase is skipped, the messages shown in the following sequence diagram SHALL be exchanged, as the corresponding resources may have changed in the meantime:

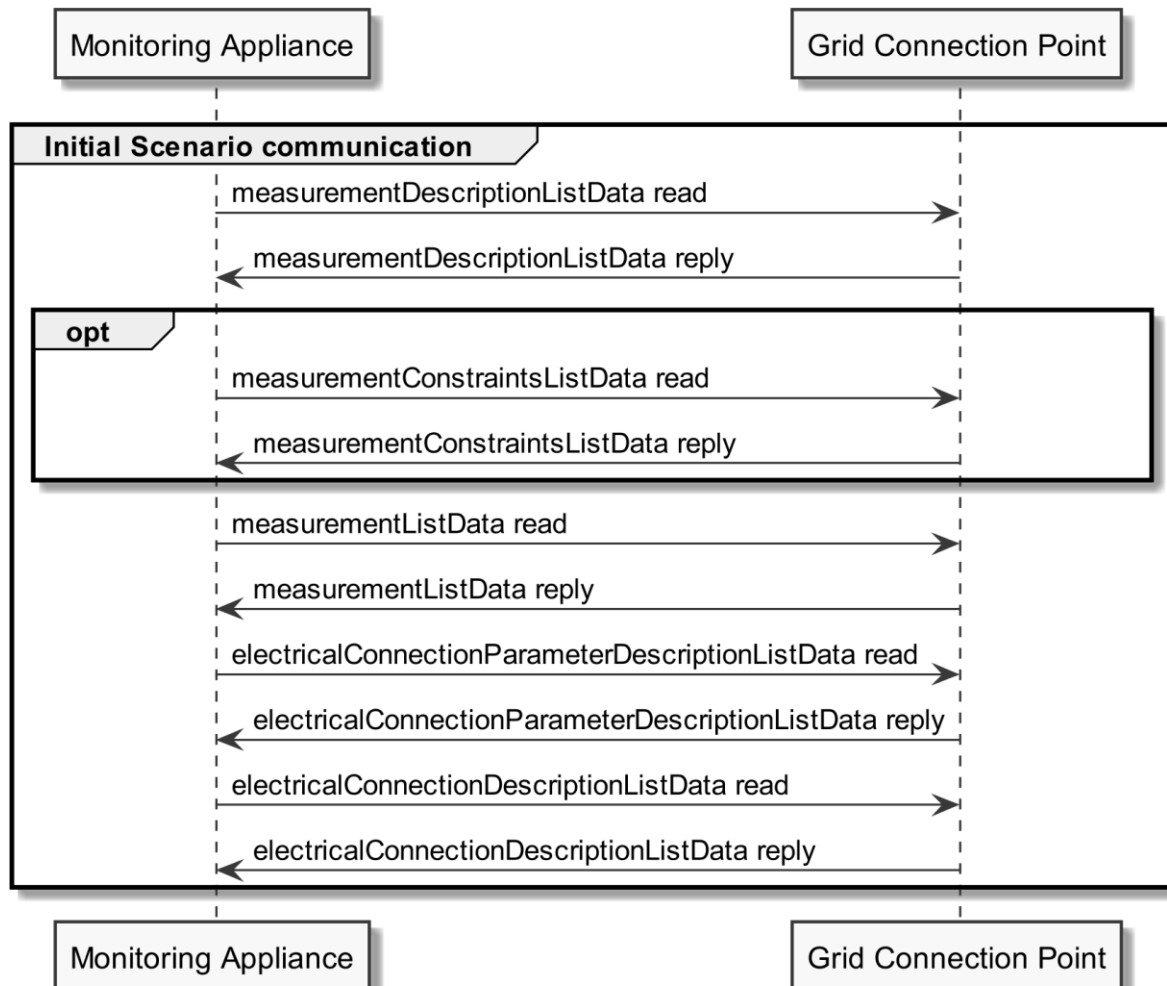


Figure 18: Scenario 5 - Initial Scenario communication sequence diagram

Note: The initiation of the optional sequence part (marked with "opt" in the figure) is optional for the "Monitoring Appliance" even if the "Grid Connection Point" can provide the requested Function.

The `measurementDescriptionListData read` SHOULD be a "partial" read operation with the following Selectors:

- `scopeType = "acCurrent"`

The `measurementConstraintsListData read`, `measurementListData read` and `electricalConnectionParameterDescriptionListData read` SHOULD be a "partial" read operation with the following Selectors:

- `measurementId` (derived from the `measurementDescriptionListData reply`)

The `electricalConnectionDescriptionListData read` SHOULD be a "partial" read operation with the following Selectors:

1475 - electricalConnectionId (derived from the electricalConnectionParameterDescriptionListData
1476 reply)

1477 Note: If partial read is not supported a full read SHALL be performed.

1478

1479 The following table shows where the required content of the messages of the sequence diagram is
1480 described:

Message name from sequence diagram	Content description in table	Scenario number in table
measurementDescriptionListData reply	Table 25	5
measurementConstraintsListData reply	Table 26	5
measurementListData reply	Table 27	5
electricalConnectionParameterDescriptionListData reply	Table 29	5
electricalConnectionDescriptionListData reply	Table 28	5

1481 *Table 38: Initial Scenario communication content references for Scenario 5*

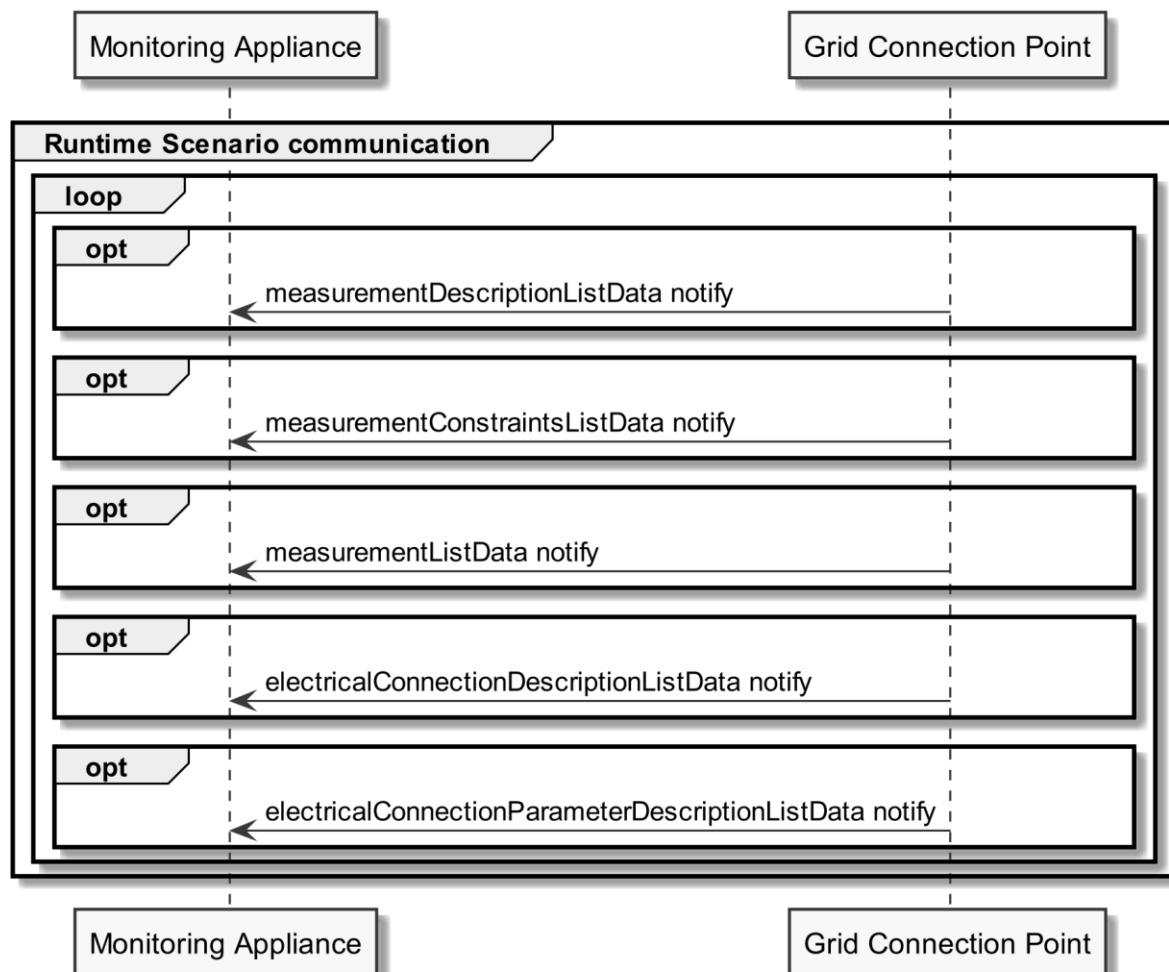
1482 Note: Within the Initial Scenario communication, the content required by this Scenario MAY not be
1483 provided completely, but later during Runtime Scenario communication.

1484

1485 **3.4.5.3 Runtime Scenario communication**

1486 Based on the Initial Scenario communication, the Runtime Scenario communication provides updates
1487 during runtime.

1488 If one of the referenced server Functions' data change the server SHALL submit the change as shown
1489 in the following figure:



1490

1491 *Figure 19: Scenario 5 - Runtime Scenario communication sequence diagram*

1492 Note: Normally, in this Scenario only the "measurementListData" Function changes during runtime.
 1493 Hence, usually no notifications of the other Functions of this Scenario are sent during runtime.

1494 Partial notifications without Selectors or Elements SHALL be supported for all Functions used in this
 1495 Scenario.

1496 For measurementDescriptionListData notify, measurementConstraintsListData notify and
 1497 measurementListData notify "partial" delete notifications SHOULD be supported with the Selector:

1498 - measurementId

1499 For electricalConnectionParameterDescriptionListData notify "partial" delete notifications SHOULD
 1500 be supported with the Selectors:

1501 - electricalConnectionId

1502 - parameterId

1503 - measurementId

1504 Note: To interpret partial notification messages correctly, the information obtained during the Initial
 1505 Scenario communication phase is required.

Note: A read operation ("polling") on all Functions is possible at any time, e.g. if a notification could not be evaluated.

The following table shows where the required content of the messages of the sequence diagram is described:

Message name from sequence diagram	Content description in table	Scenario number in table
measurementDescriptionListData notify	Table 25	5
measurementConstraintsListData notify	Table 26	5
measurementListData notify	Table 27	5
electricalConnectionParameterDescriptionListData notify	Table 29	5
electricalConnectionDescriptionListData notify	Table 28	5

Table 39: Runtime Scenario communication content references for Scenario 5

3.4.5.4 Additional information

Note: Both, consumed and produced current, are represented by the single measurement value of this Scenario. Whether the device consumes or produces current is indicated by the sign of the measurement value: The Element "positiveEnergyDirection" in the Function "electricalConnectionDescriptionListData" determines how the sign SHALL be interpreted. Within this Use Case the Element has a fixed value of "consume". Hence, the "passive sign convention" is applied and therefore positive values SHALL be interpreted as consumed current and negative values SHALL be interpreted as produced current (see section 2.6.1 for details).

Note: Within this Use Case, only the newest measurement value SHALL be stated ([MGCP-004]). Additional historical values are forbidden.

3.4.6 Scenario 6 - Monitor voltage phase details

3.4.6.1 Pre-Scenario communication

- Detailed discovery:** Actors that act as client within this Scenario, need to know the addresses of the server Features used in the Initial Scenario communication. If the address of a particular server Feature is not known, the detailed discovery must be used, as described in section 3.3.2.
- Binding:** Binding SHOULD NOT be used for this Scenario.
- Subscription:** Actors SHALL create a subscription for each server Feature that is relevant for the corresponding Actor within this Scenario, as described in section 3.3.4.

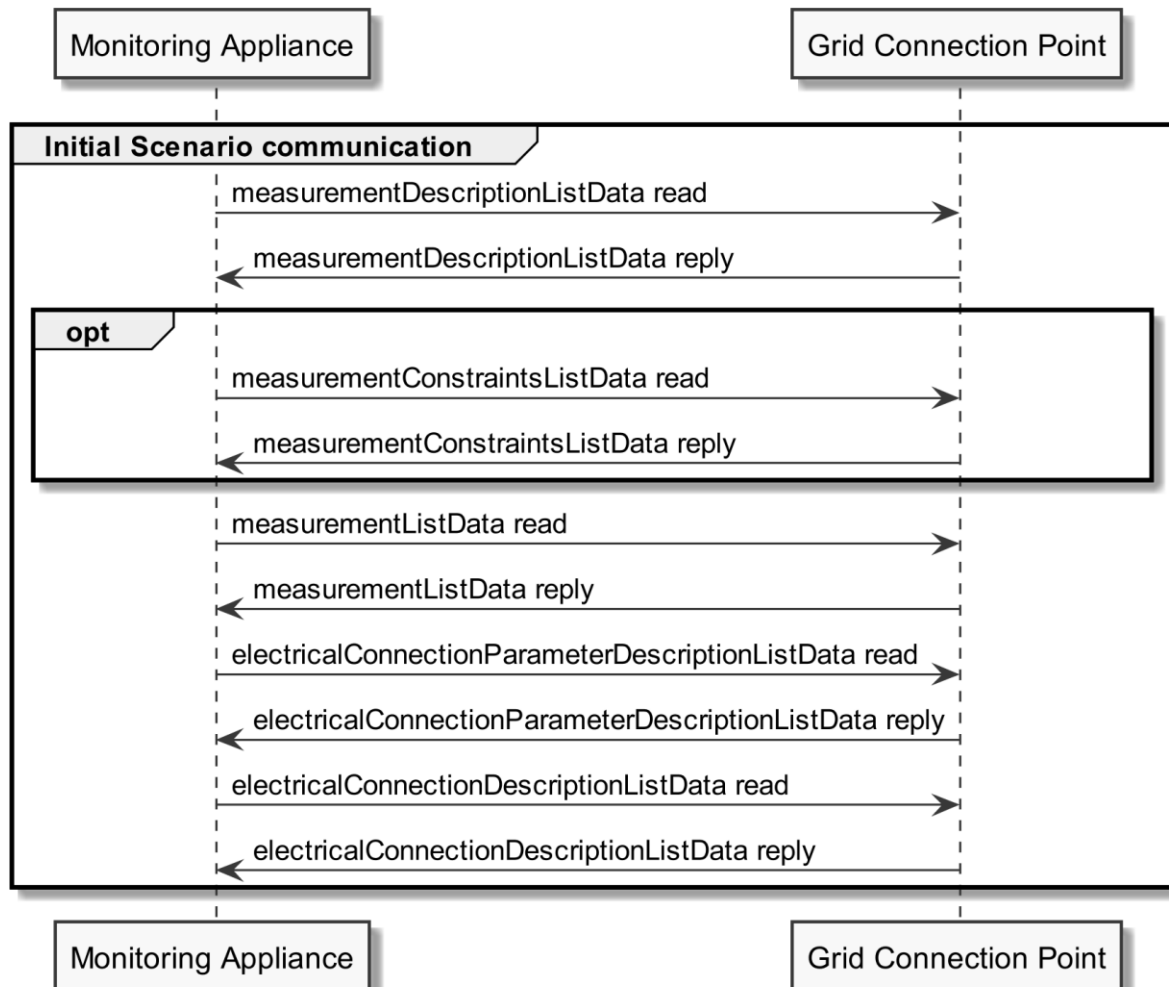
The Initial Scenario communication SHALL start at the latest when the required resources on an Actor are known and the necessary binding and subscription procedures have been finished. However, as soon as the address of a required resource is known, the Initial Scenario communication for this resource MAY start already, even if the addresses of other required resources are not known yet.

1537 If required resources are removed and added again, they are re-discovered, and the Initial Scenario
 1538 communication is triggered again for those resources.

1539

1540 **3.4.6.2 Initial Scenario communication**

1541 Each time a (re-)connection is established, even if the Pre-Scenario communication phase is skipped,
 1542 the messages shown in the following sequence diagram SHALL be exchanged, as the corresponding
 1543 resources may have changed in the meantime:



1544

1545 *Figure 20: Scenario 6 - Initial Scenario communication sequence diagram*

1546 The measurementDescriptionListData read SHOULD be a "partial" read operation with the following
 1547 Selector:

- 1548 - scopeType = "acVoltage"

1549 The measurementConstraintsListData read, measurementListData read and
 1550 electricalConnectionParameterDescriptionListData read SHOULD be "partial" read operations with
 1551 the following Selector:

- 1552 - measurementId (derived from the measurementDescriptionListData reply)

1553 The electricalConnectionDescriptionListData read SHOULD be a "partial" read operation with the
 1554 following Selector:

- 1555 - electricalConnectionId (derived from the electricalConnectionParameterDescriptionListData
 1556 reply)

1557 Note: If partial read is not supported, a full read SHALL be performed.

1558

1559 The following table shows where the required content of the messages from the sequence diagram is
 1560 described:

Message name from sequence diagram	Content description in table	Scenario number in table
measurementDescriptionListData reply	Table 25	6
measurementConstraintsListData reply	Table 26	6
measurementListData reply	Table 27	6
electricalConnectionParameterDescriptionListData reply	Table 29	6
electricalConnectionDescriptionListData reply	Table 28	6

1561 *Table 40: Initial Scenario communication content references for Scenario 6*

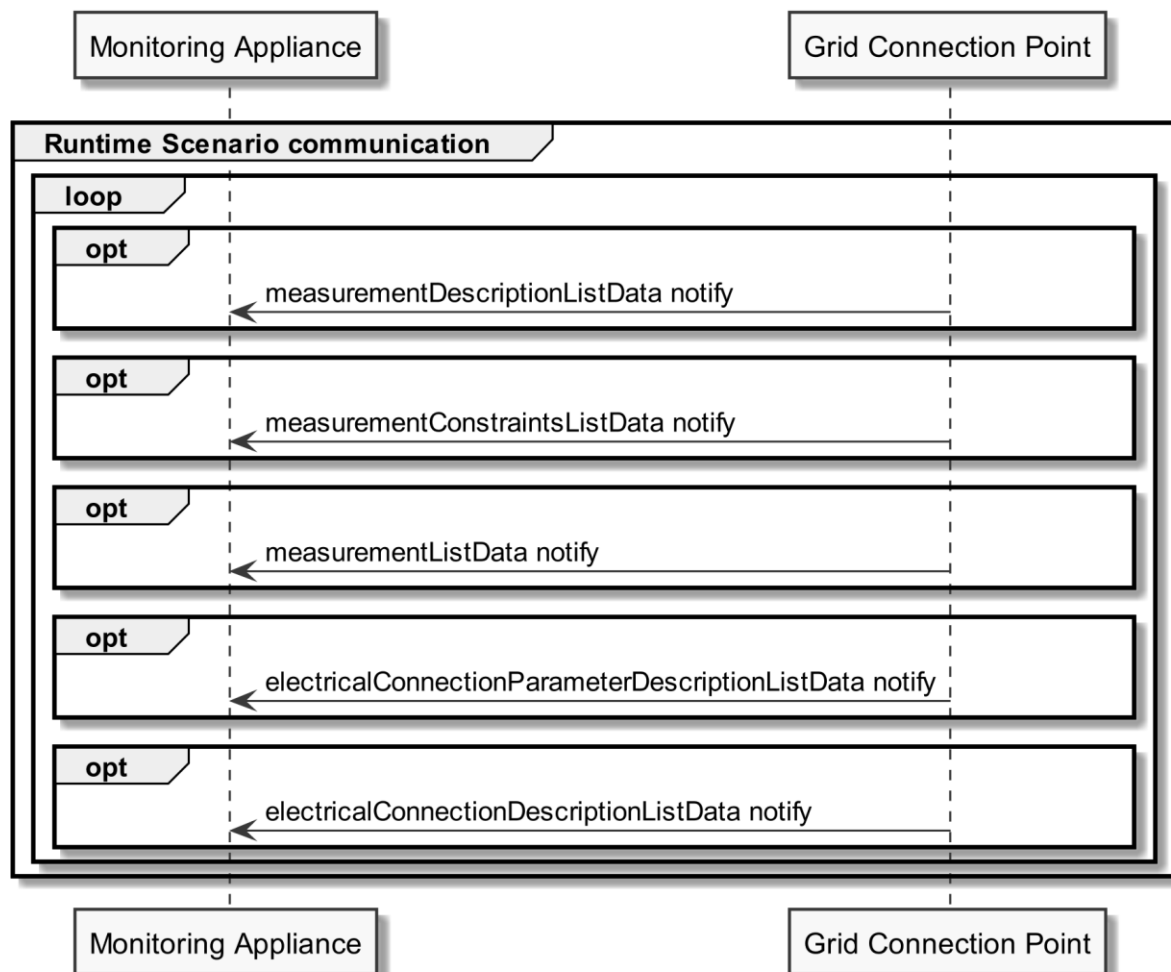
1562 Note: Within the Initial Scenario communication, the content required by this Scenario MAY not be
 1563 provided completely, but later during Runtime Scenario communication.

1564

1565 **3.4.6.3 Runtime Scenario communication**

1566 Based on the Initial Scenario communication, the Runtime Scenario communication provides updates
 1567 during runtime.

1568 If one of the referenced server Functions' data change, the server SHALL submit the change as shown
 1569 in the following figure:



1570

1571 *Figure 21: Scenario 6 - Runtime Scenario communication sequence diagram*

1572 Note: Normally, in this Scenario only the "measurementListData" Function changes during runtime.
 1573 Hence, usually no notifications of the other Functions of this Scenario are sent during runtime.

1574 Partial notifications without Selectors or Elements SHALL be supported for all Functions used in this
 1575 Scenario.

1576 For measurementDescriptionListData notify, measurementConstraintsListData notify and
 1577 measurementListData notify "partial" delete notifications SHOULD be supported with the Selector:

1578 - measurementId

1579 For electricalConnectionParameterDescriptionListData notify "partial" delete notifications SHOULD
 1580 be supported with the Selectors:

1581 - electricalConnectionId

1582 - parameterId

1583 - measurementId

1584 Note: To interpret partial notification messages correctly, the information obtained during the Initial
 1585 Scenario communication phase is required.

Note: A read operation ("polling") on all Functions is possible at any time, e.g. if a notification could not be evaluated.

The following table shows where the required content of the messages of the sequence diagram is described:

Message name from sequence diagram	Content description in table	Scenario number in table
measurementDescriptionListData notify	Table 25	6
measurementConstraintsListData notify	Table 26	6
measurementListData notify	Table 27	6
electricalConnectionParameterDescriptionListData notify	Table 29	6
electricalConnectionDescriptionListData notify	Table 28	6

Table 41: Runtime Scenario communication content references for Scenario 6

3.4.6.4 Additional information

None.

3.4.7 Scenario 7 - Monitor frequency

3.4.7.1 Pre-Scenario communication

- Detailed discovery:** Actors that act as client within this Scenario, need to know the addresses of the server Features used in the Initial Scenario communication. If the address of a particular server Feature is not known, the detailed discovery must be used, as described in section 3.3.2.
- Binding:** Binding SHOULD NOT be used for this Scenario.
- Subscription:** Actors SHALL create a subscription for each server Feature that is relevant for the corresponding Actor within this Scenario, as described in section 3.3.4.

The Initial Scenario communication SHALL start at the latest when the required resources on an Actor are known and the necessary binding and subscription procedures have been finished. However, as soon as the address of a required resource is known, the Initial Scenario communication for this resource MAY start already, even if the addresses of other required resources are not known yet.

If required resources are removed and added again, they are re-discovered, and the Initial Scenario communication is triggered again for those resources.

3.4.7.2 Initial Scenario communication

Each time a (re-)connection is established, even if the Pre-Scenario communication phase is skipped, the messages shown in the following sequence diagram SHALL be exchanged, as the corresponding resources may have changed in the meantime:

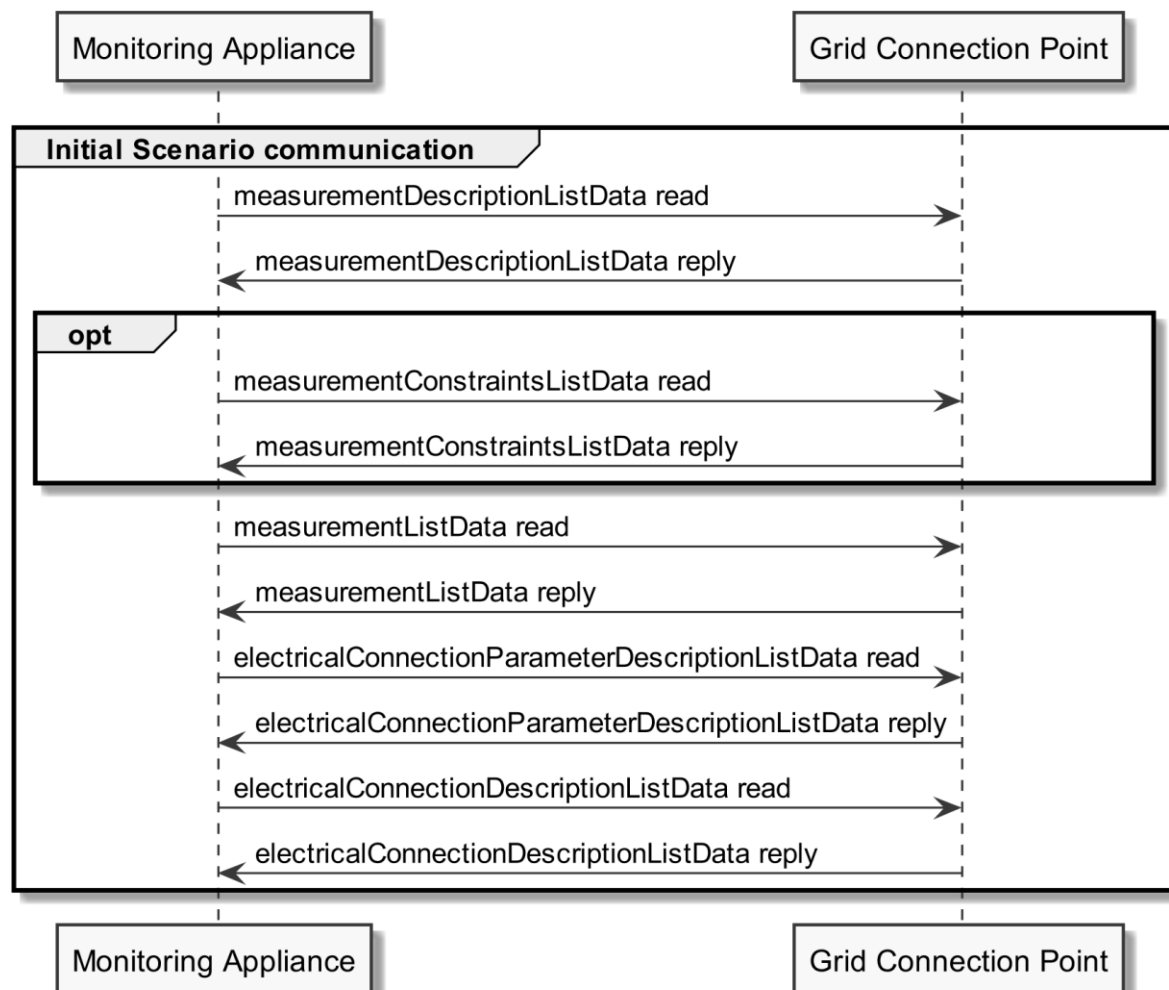


Figure 22: Scenario 7 - Initial Scenario communication sequence diagram

The `measurementDescriptionListData read` SHOULD be a "partial" read operation with the following Selector:

- `scopeType = "acFrequency"`

The `measurementConstraintsListData read`, `measurementListData read` and `electricalConnectionParameterDescriptionListData read` SHOULD be "partial" read operations with the following Selector:

- `measurementId` (derived from the `measurementDescriptionListData reply`)

The `electricalConnectionDescriptionListData read` SHOULD be a "partial" read operation with the following Selector:

- `electricalConnectionId` (derived from the `electricalConnectionParameterDescriptionListData reply`)

Note: If partial read is not supported, a full read SHALL be performed.

1631 The following table shows where the required content of the messages from the sequence diagram is
 1632 described:

Message name from sequence diagram	Content description in table	Scenario number in table
measurementDescriptionListData reply	Table 25	7
measurementConstraintsListData reply	Table 26	7
measurementListData reply	Table 27	7
electricalConnectionParameterDescriptionListData reply	Table 29	7
electricalConnectionDescriptionListData reply	Table 28	7

1633 *Table 42: Initial Scenario communication content references for Scenario 7*

1634 Note: Within the Initial Scenario communication, the content required by this Scenario MAY not be
 1635 provided completely, but later during Runtime Scenario communication.

1636

1637 **3.4.7.3 Runtime Scenario communication**

1638 Based on the Initial Scenario communication, the Runtime Scenario communication provides updates
 1639 during runtime.

1640 If one of the referenced server Functions' data change, the server SHALL submit the change as shown
 1641 in the following figure:

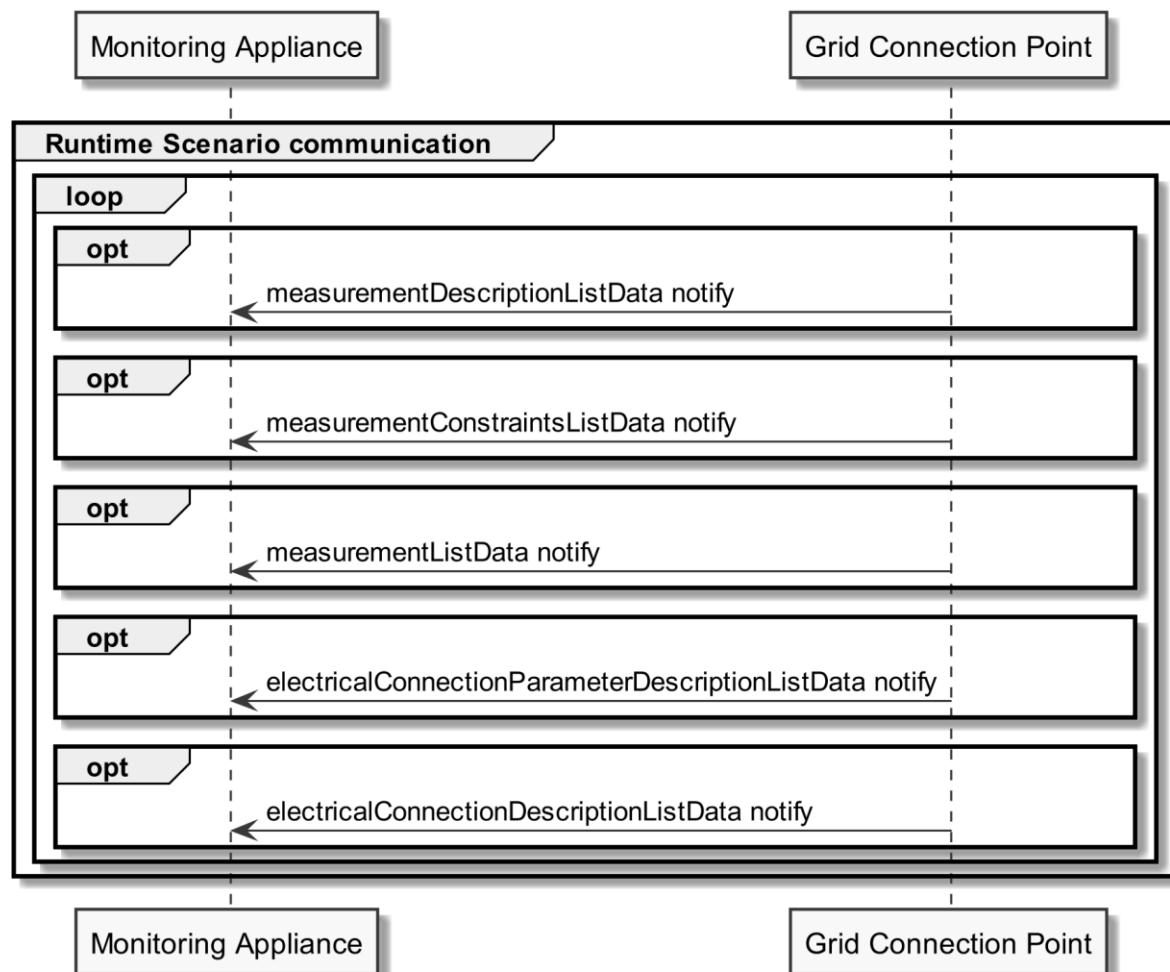


Figure 23: Scenario 7 - Runtime Scenario communication sequence diagram

Note: Normally, in this Scenario only the "measurementListData" Function changes during runtime. Hence, usually no notifications of the other Functions of this Scenario are sent during runtime.

Partial notifications without Selectors or Elements SHALL be supported for all Functions used in this Scenario.

For measurementDescriptionListData notify, measurementConstraintsListData notify and measurementListData notify "partial" delete notifications SHOULD be supported with the Selector:

- measurementId

For electricalConnectionParameterDescriptionListData notify "partial" delete notifications SHOULD be supported with the Selectors:

- electricalConnectionId
- parameterId
- measurementId

Note: To interpret partial notification messages correctly, the information obtained during the Initial Scenario communication phase is required.

1658 Note: A read operation ("polling") on all Functions is possible at any time, e.g. if a notification could
 1659 not be evaluated.

1660

1661 The following table shows where the required content of the messages of the sequence diagram is
 1662 described:

Message name from sequence diagram	Content description in table	Scenario number in table
measurementDescriptionListData notify	Table 25	7
measurementConstraintsListData notify	Table 26	7
measurementListData notify	Table 27	7
electricalConnectionParameterDescriptionListData notify	Table 29	7
electricalConnectionDescriptionListData notify	Table 28	7

1663 *Table 43: Runtime Scenario communication content references for Scenario 7*

1664

1665 **3.4.7.4 Additional information**

1666 None.

1667